

SYMMETRIC KEY ENCRYPTION

Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

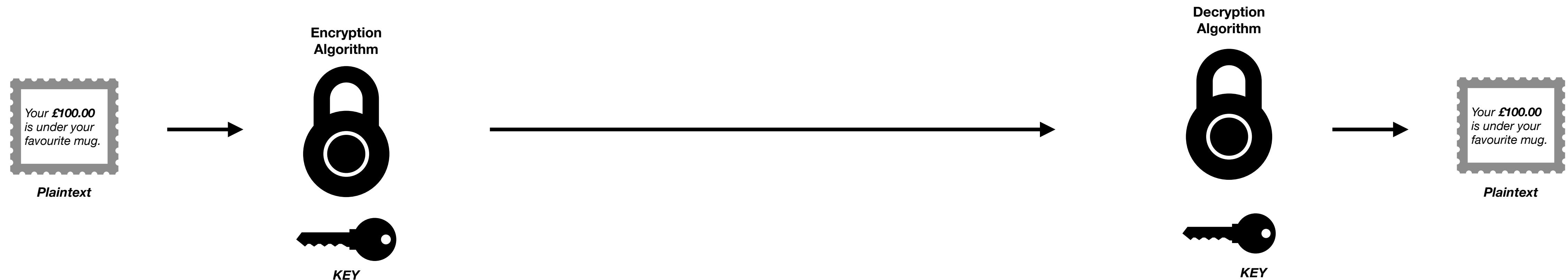
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Cryptography

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Data encrypted by Bob with a key can only be decrypted by Alice using that same key. Because the key must be kept private (confidential), symmetric encryption is also called private key cryptography.

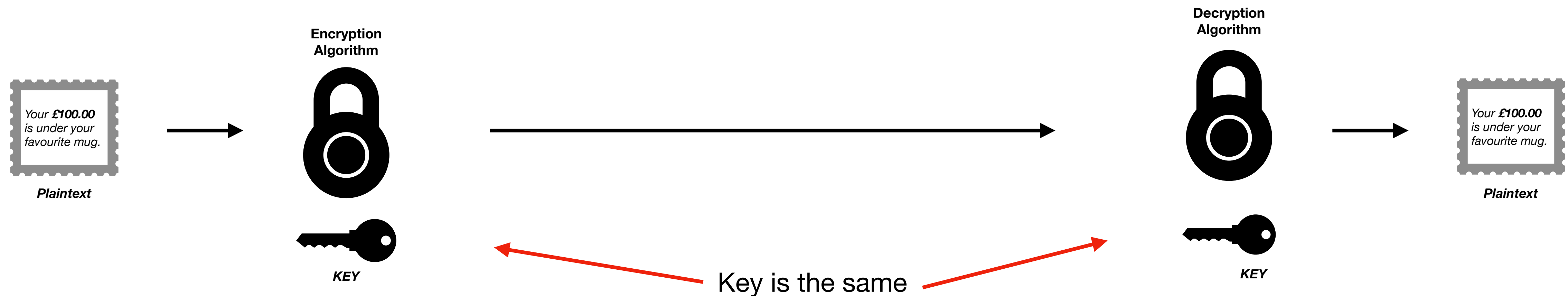


Cryptography

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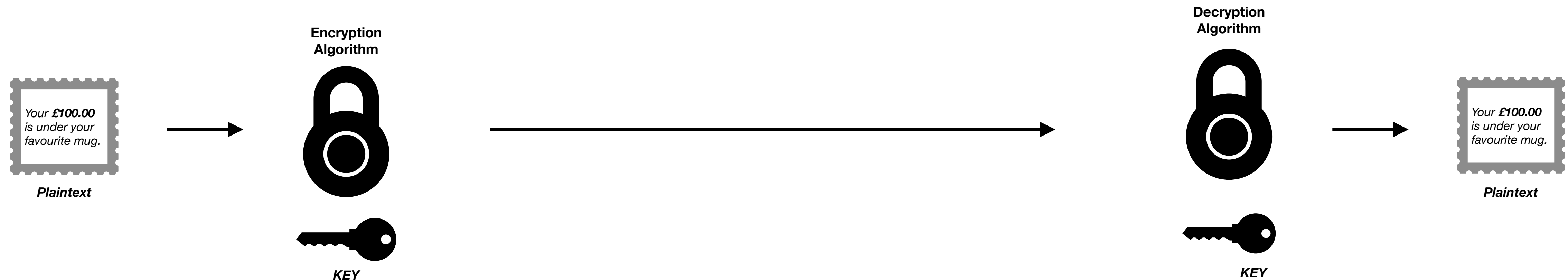


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ADVANTAGES

- Fast and convenient to set up.
- Requires less resources compared to Asymmetric encryption

DISADVANTAGES

- Sharing the key
- Lots of keys to manage
- More damage if the keys are compromised
- Cannot provide digital signatures that cannot be repudiated.

Cryptography

CRYPTOGRAPHIC ALGORITHMS - ONE-TIME PAD

One-time pad (OTP) combines plaintext with a random key.

A **pad** is a long sequence of random letters. The letters are combined with the plaintext message to produce the ciphertext. To decipher the message, the recipient must have a copy of the pad to reverse the process.

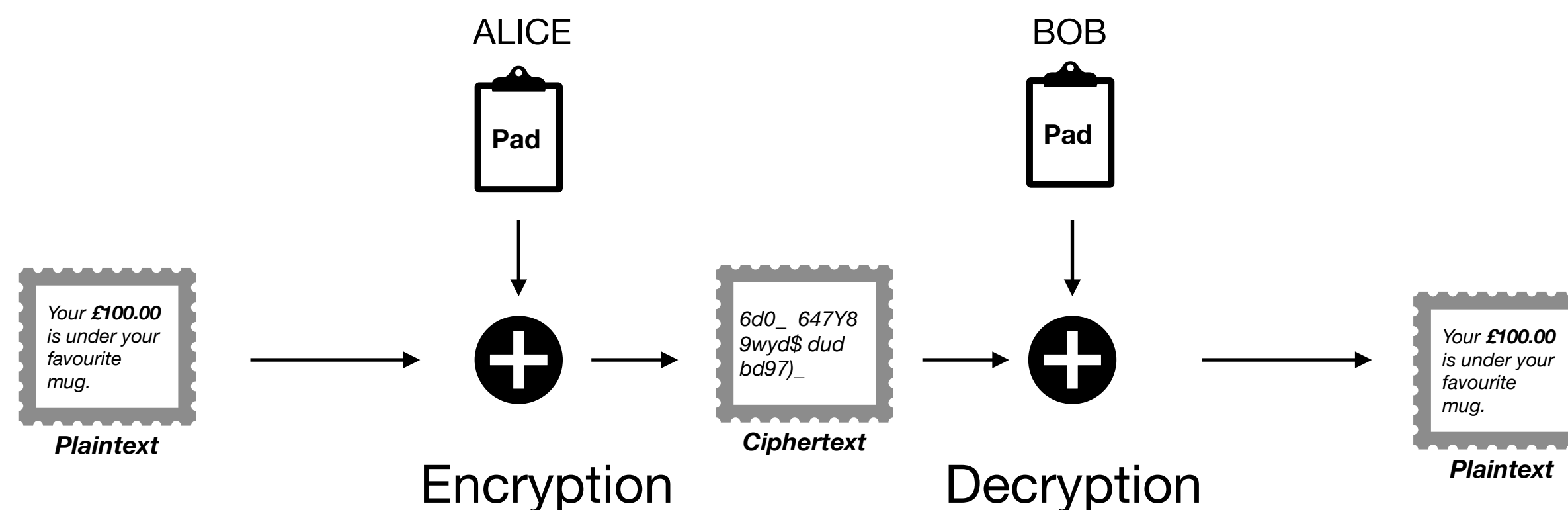
To encipher a message, the position in the alphabet of the first letter in the plaintext message is added to the position in the alphabet of the first random letter from the pad. To decipher a message, the recipient takes the first letter of the ciphertext and subtracts the first random letter from the pad (any negative numbers are wrapped around to the end of the alphabet).

To encrypt the message

$$M \oplus K = M'$$

To decrypt the message

$$M' \oplus K = M$$



Cryptography

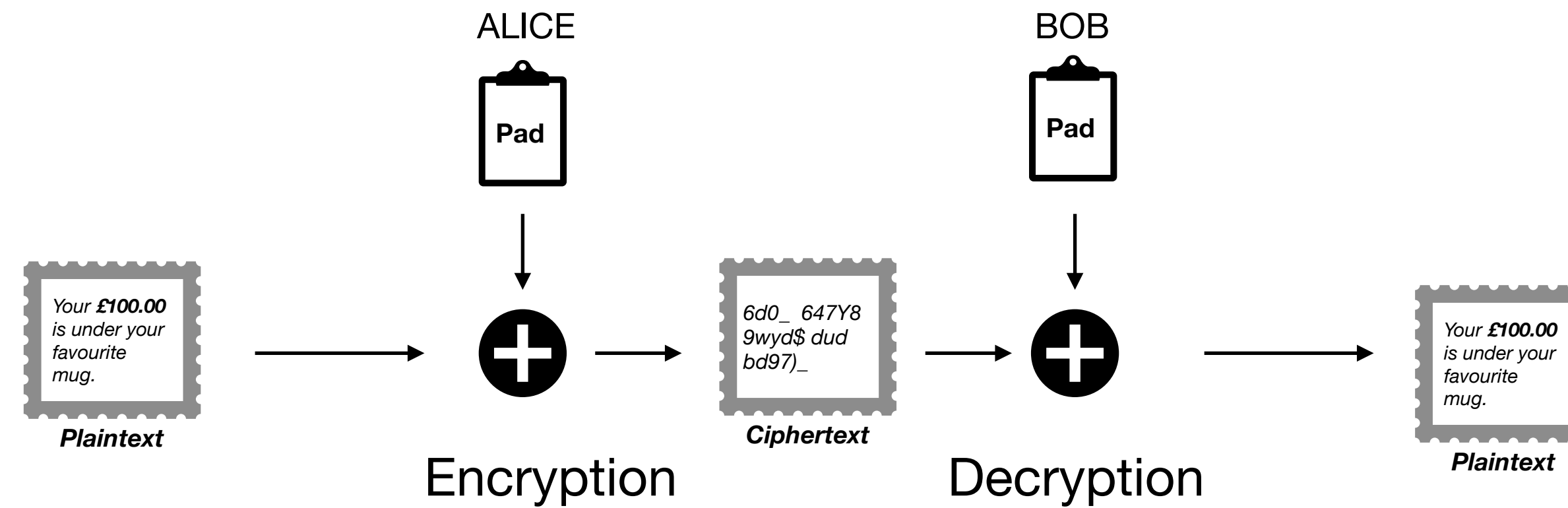
CRYPTOGRAPHIC ALGORITHMS - ONE-TIME PAD

To encrypt the message

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- Perfect cipher
- The key is only used once and destroyed after
- Key is the same length as the message

Cryptography

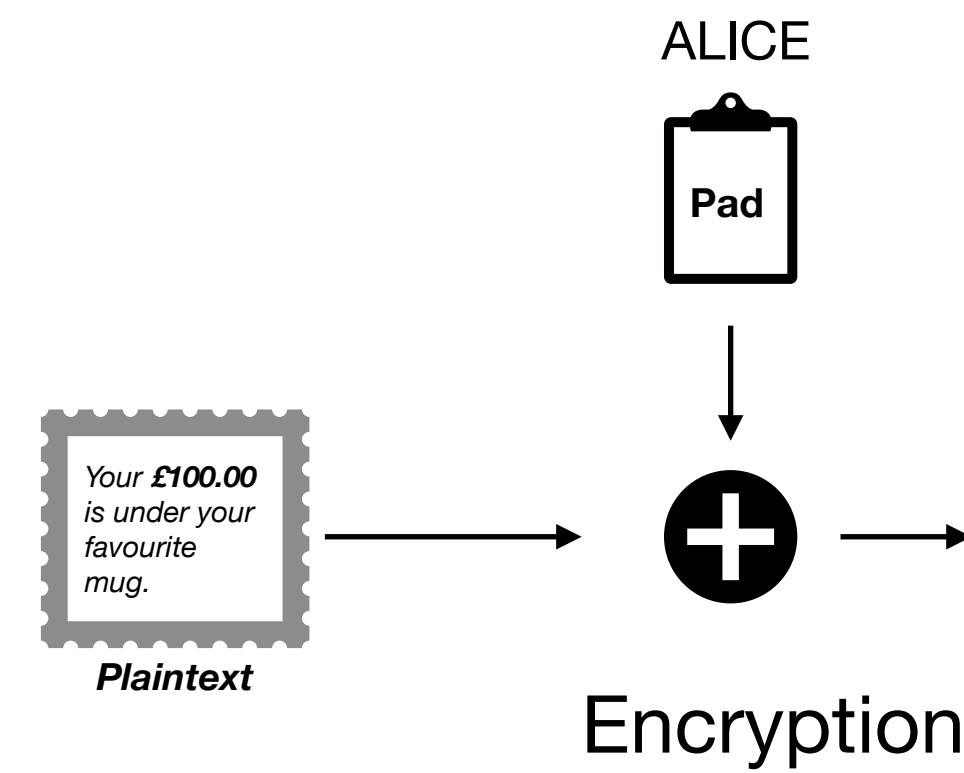
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Message = H e l l o

Key = K * e 0 !



Cryptography

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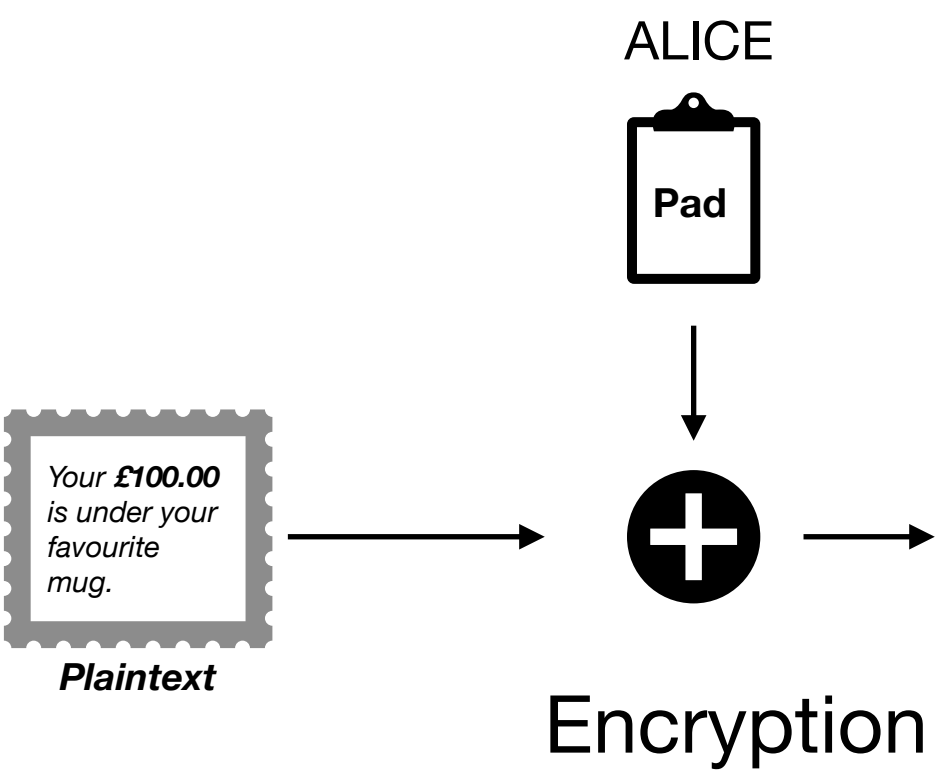
$$M \oplus K = M'$$

Message = H e l l o

Key = K * e 0 !

H	72
e	101
l	108
l	108
o	111

K	75
*	42
e	101
0	48
!	33



Cryptography

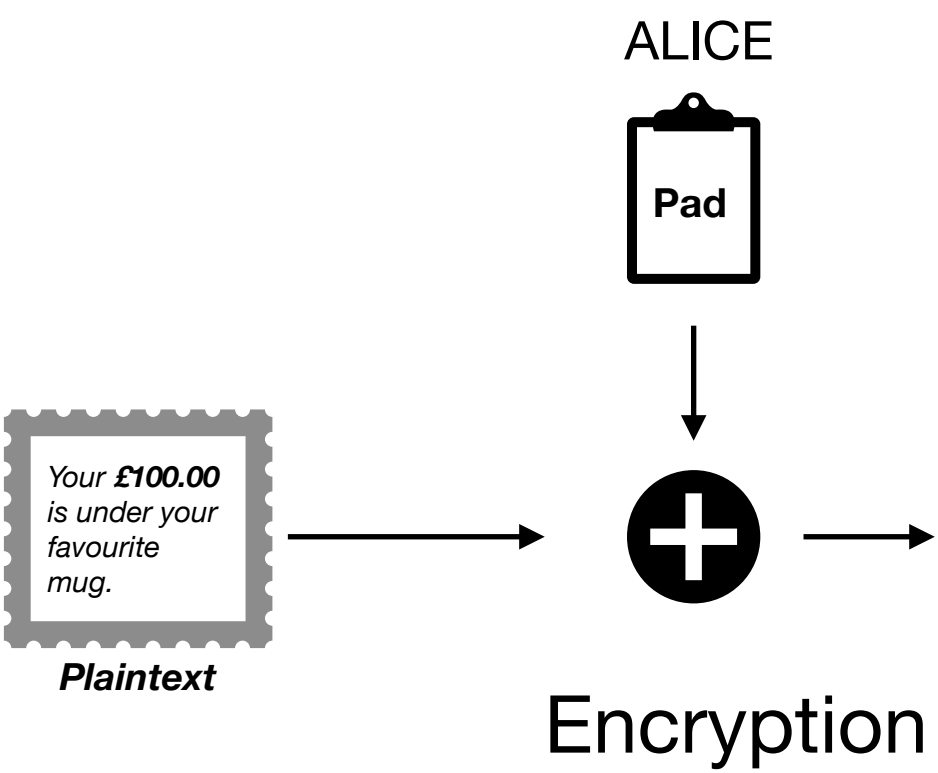
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Convert to binary

H	72	H	1001000
e	101	e	1100101
l	108	l	1101100
l	108	l	1101100
o	111	o	1101111

K	75	K	1001011
*	42	*	101010
e	101	e	1100101
0	48	0	110000
!	33	!	100001

Cryptography

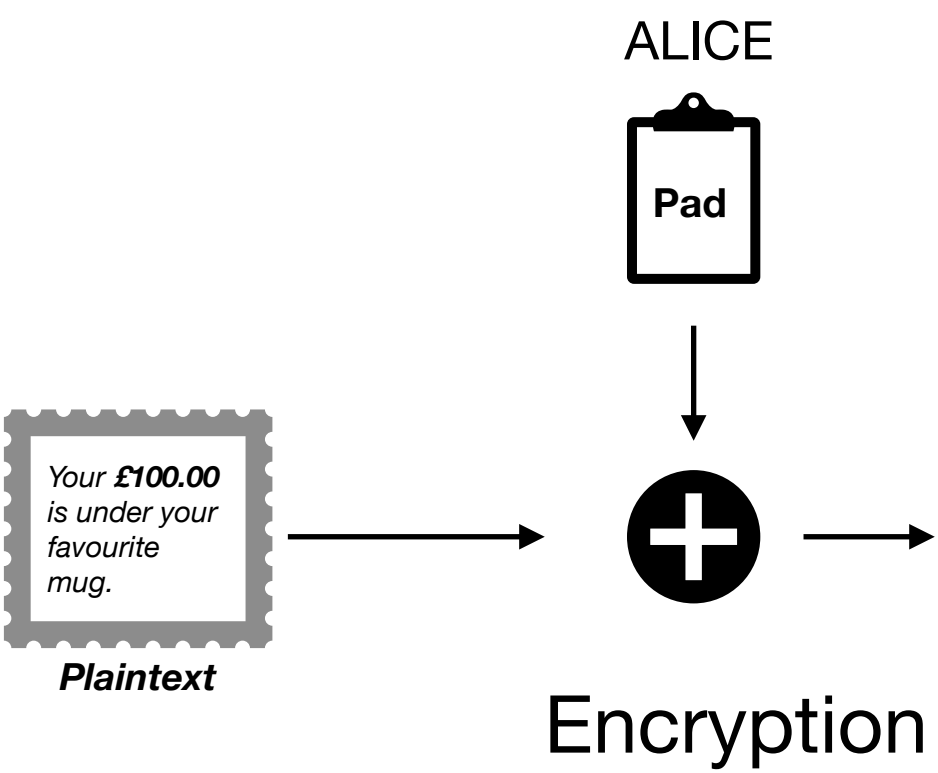
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$M \oplus K = M'$

XOR

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Cryptography

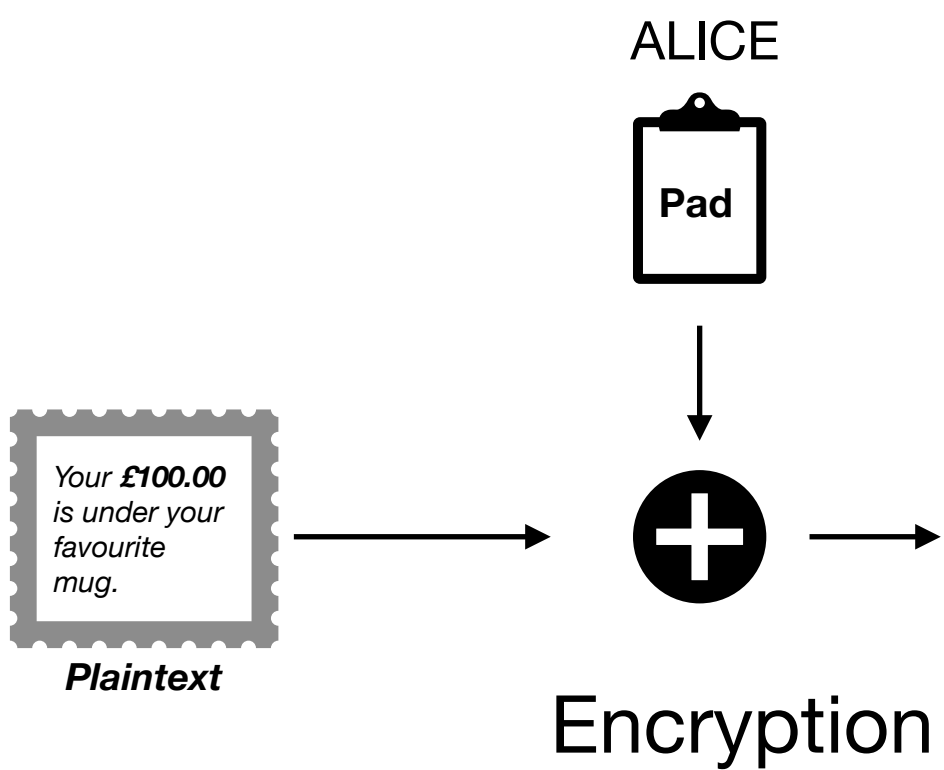
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H 0100 1000 =

e 0110 0101 =

l 0110 1100 =

l 0110 1100 =

o 0110 1111 =

Cryptography

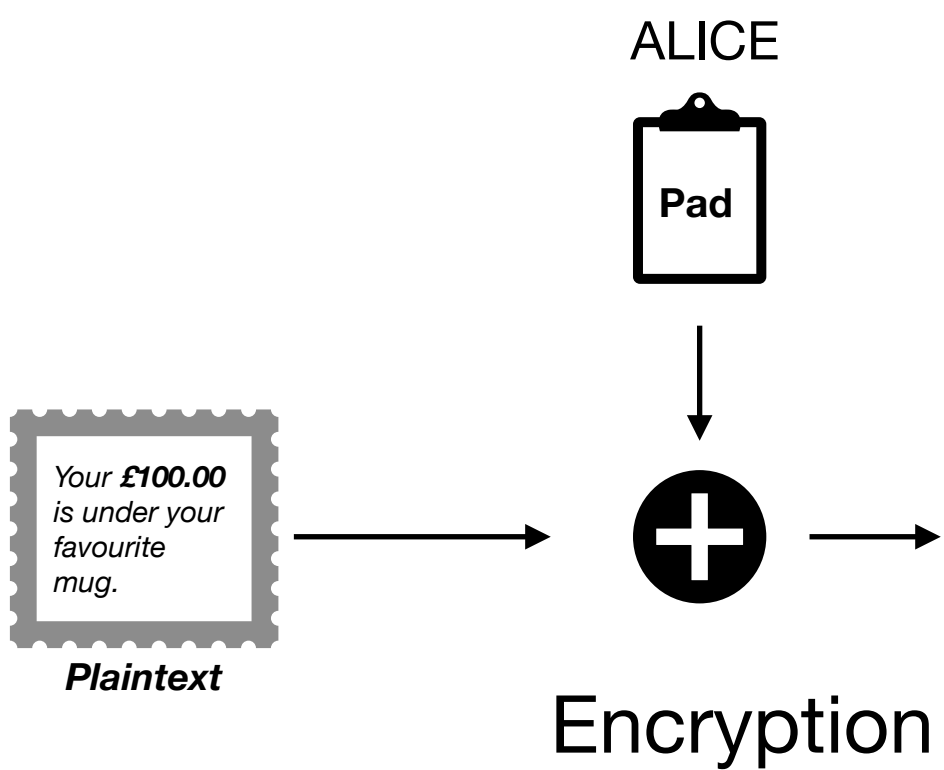
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H	0100 1000	=
K	0100 1011	
e	0110 0101	=
*	0010 1010	
l	0110 1100	=
e	0110 0101	
l	0110 1100	=
0	0011 0000	
o	0110 1111	=
!	0010 0001	

Cryptography

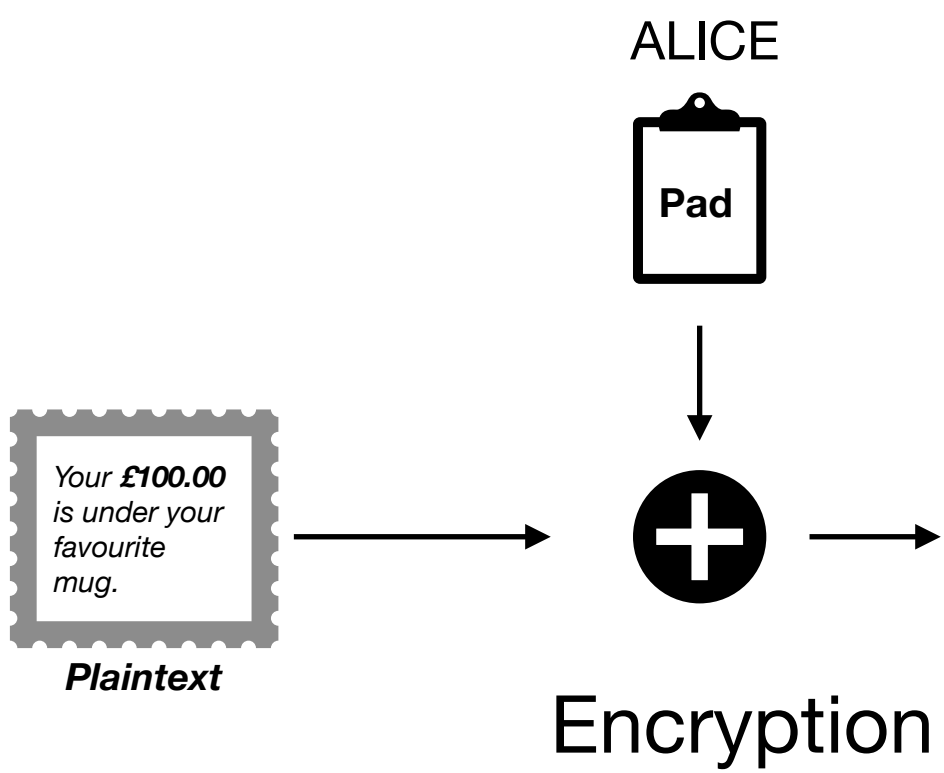
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$M \oplus K = M'$

XOR

H e l l o
K * e 0 !

BINARY

H	0100 1000	=	0000 0011
K	0100 1011		
e	0110 0101	=	0100 1111
*	0010 1010		
l	0110 1100	=	0000 1001
e	0110 0101		
l	0110 1100	=	0101 1100
0	0011 0000		
o	0110 1111	=	0100 1110
!	0010 0001		

Cryptography

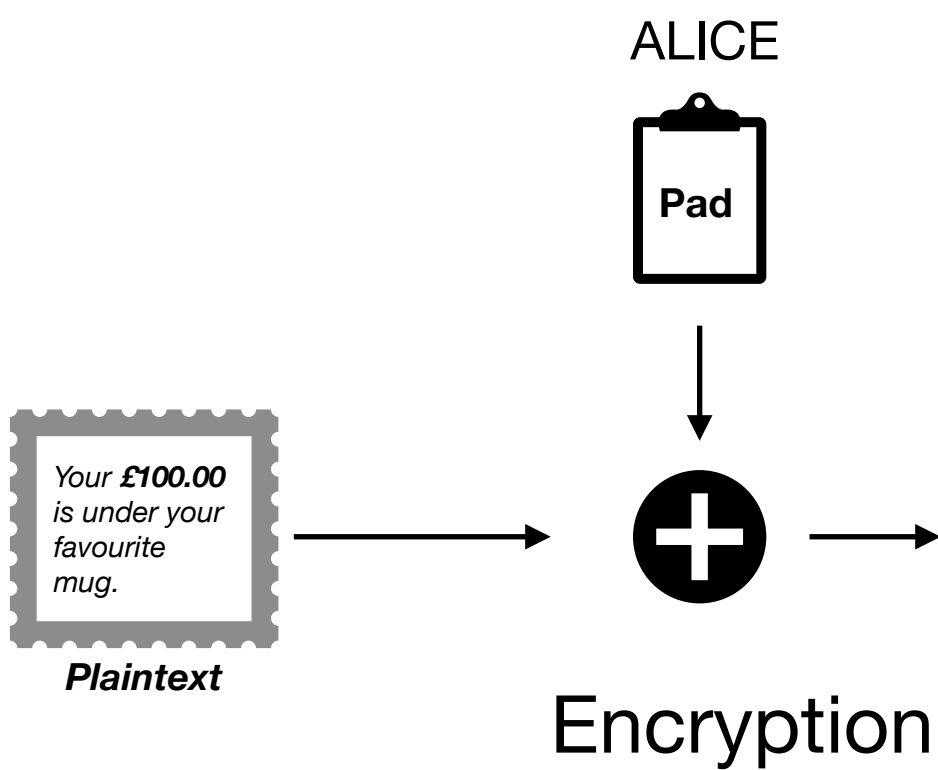
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 H e l l o

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BINARY

H 0100 1000
K 0100 1011
===== 0000 0011

e 0110 0101
* 0010 1010
===== 0100 1111

l 0110 1100
e 0110 0101
===== 0000 1001

l 0110 1100
0 0011 0000
===== 0101 1100

o 0110 1111
! 0010 0001
===== 0100 1110

Even though they are both letter *l* you get a different results.

Cryptography

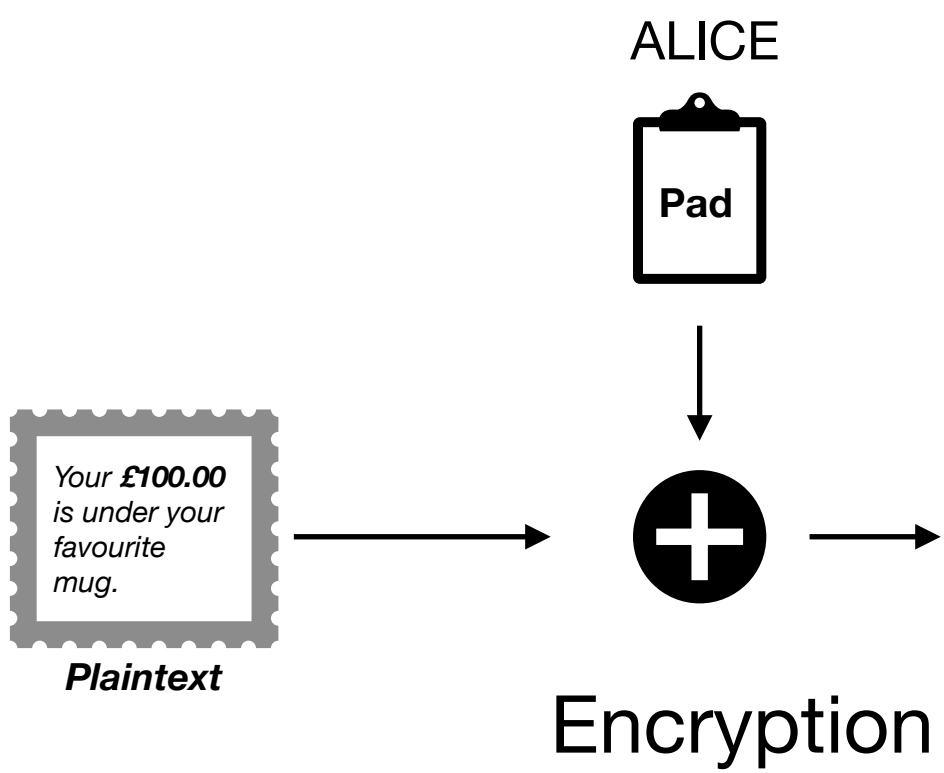
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$M \oplus K = M'$

XOR

H	e	l	l	o
K	*	e	0	!

		BINARY	DEC
H	0100 1000	= 0000 0011	3
K	0100 1011		
e	0110 0101	= 0100 1111	79
*	0010 1010		
l	0110 1100	= 0000 1001	9
e	0110 0101		
l	0110 1100	= 0101 1100	92
0	0011 0000		
o	0110 1111	= 0100 1110	78
!	0010 0001		

Cryptography

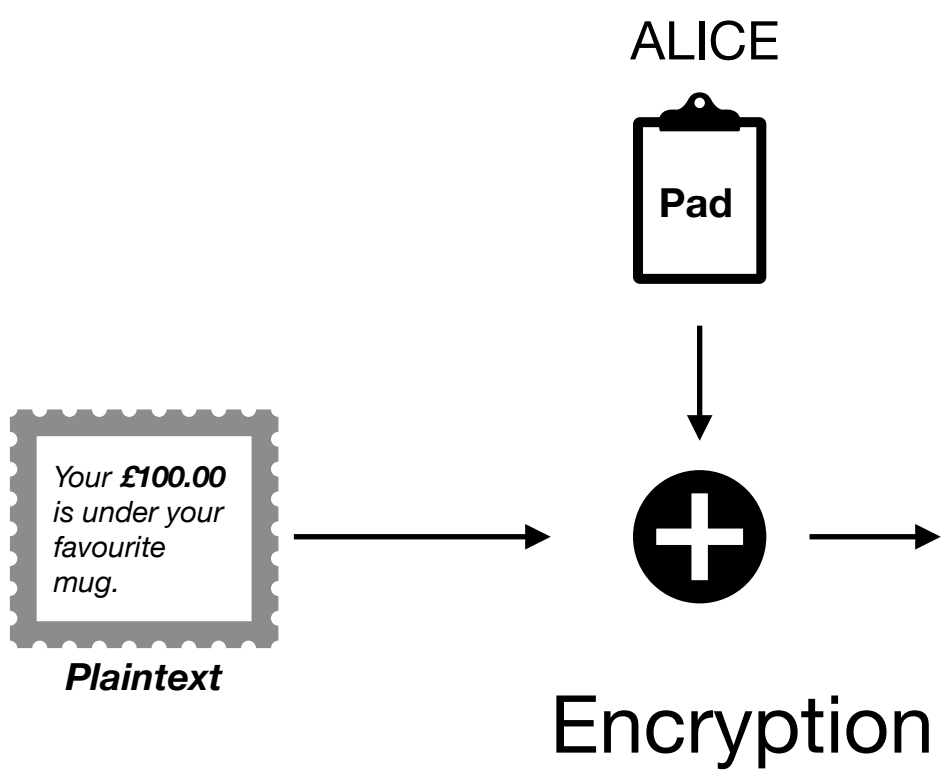
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XOR

H e l l o
K * e 0 !

		BINARY	DEC	CHAR
H	$\frac{0100\ 1000}{0100\ 1011} =$	0000 0011	3	☐
e	$\frac{0110\ 0101}{0010\ 1010} =$	0100 1111	79	O
l	$\frac{0110\ 1100}{0110\ 0101} =$	0000 1001	9	☐
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o	$\frac{0110\ 1111}{0010\ 0001} =$	0100 1110	78	N

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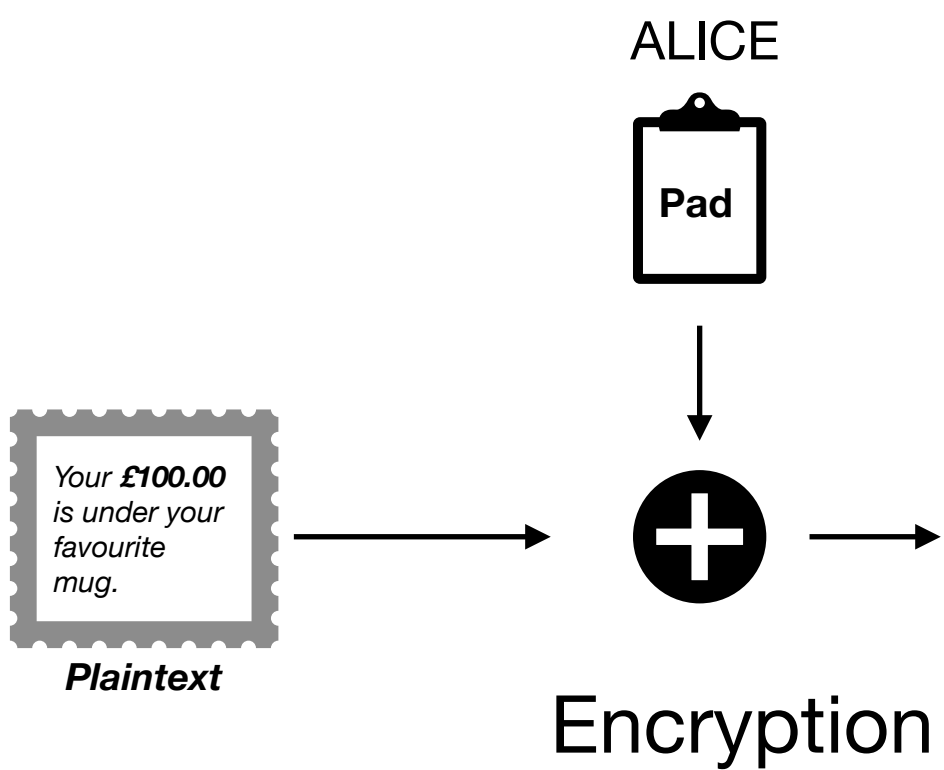
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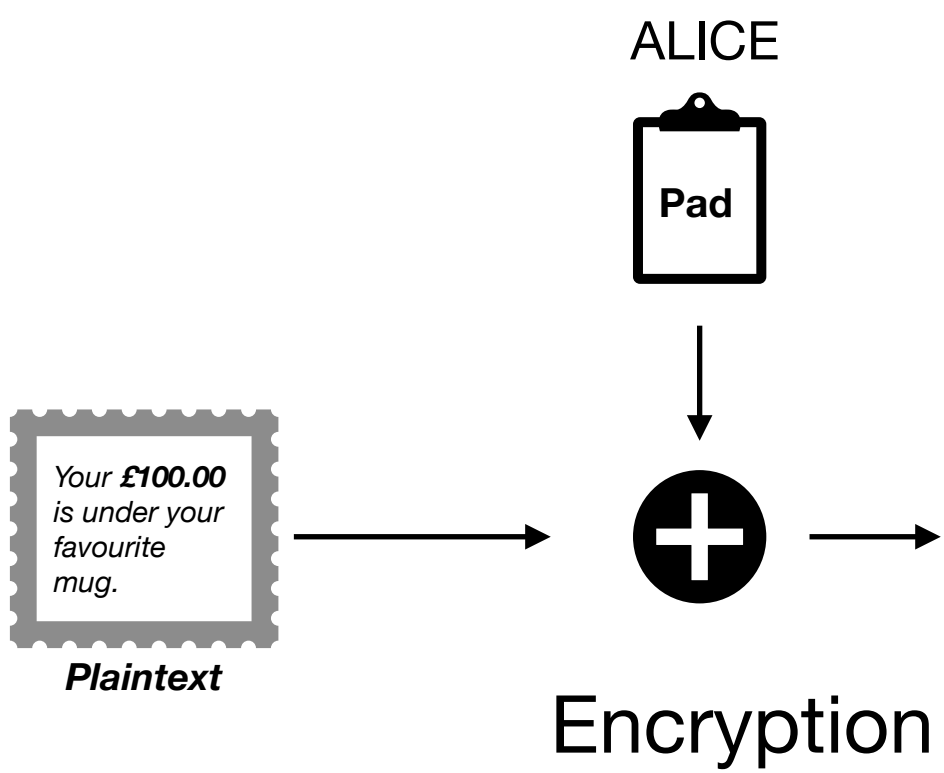
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XOR

 H e l l o

 K * e 0 !

		BINARY	DEC	CHAR	HEX
H	$\frac{0100\ 1000}{0100\ 1011} =$	0000 0011	3	☐	3
e	$\frac{0110\ 0101}{0010\ 1010} =$	0100 1111	79	O	4F
l	$\frac{0110\ 1100}{0110\ 0101} =$	0000 1001	9	☐	9
l	$\frac{0110\ 1100}{0011\ 0000} =$	0101 1100	92	\	5C
o	$\frac{0110\ 1111}{0010\ 0001} =$	0100 1110	78	N	4E

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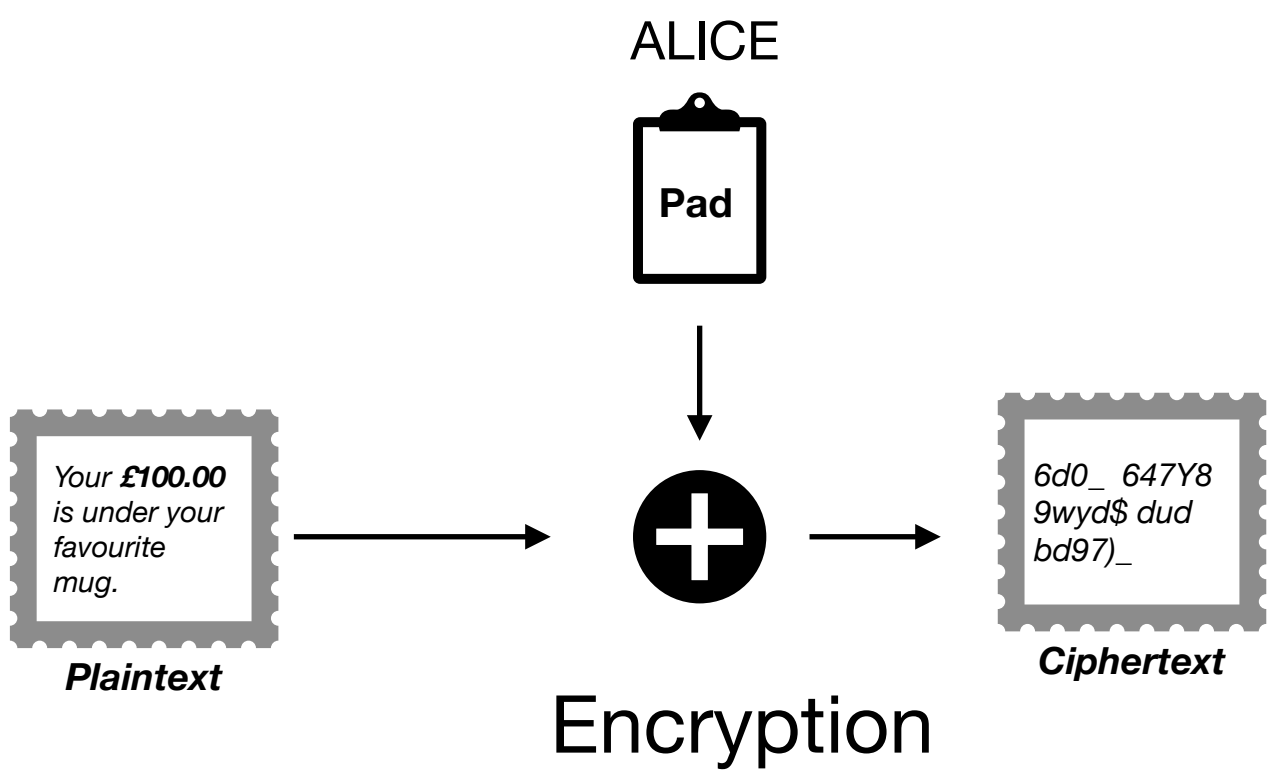
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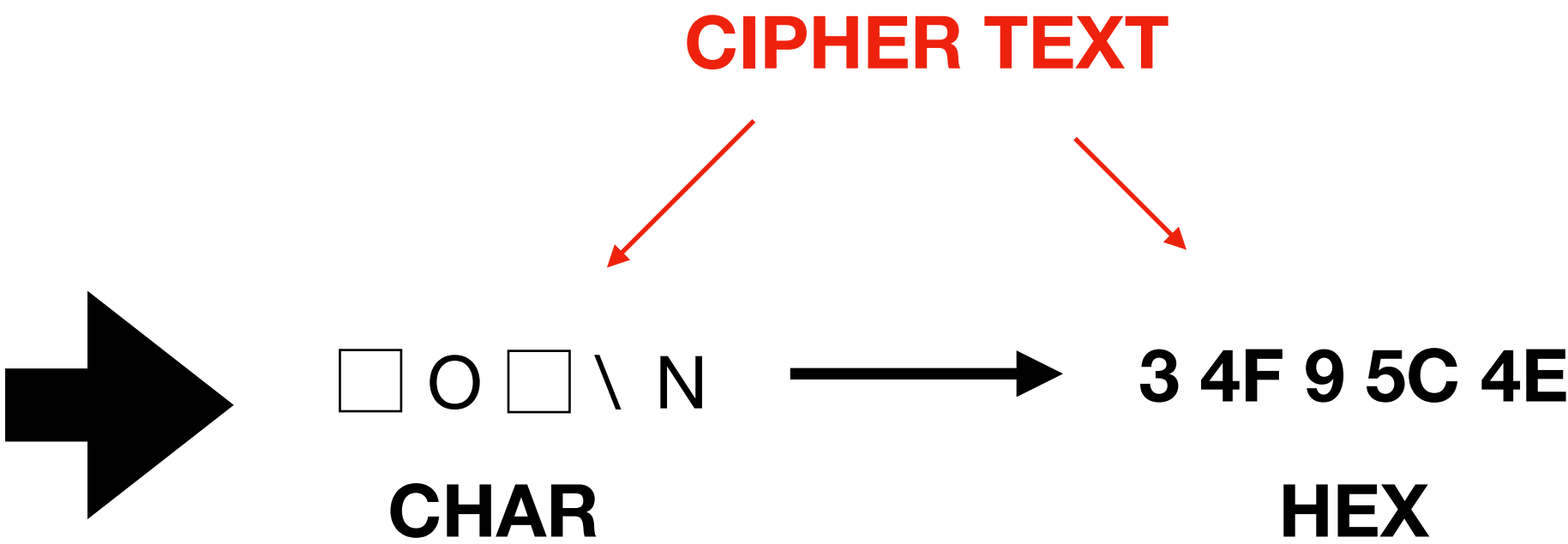
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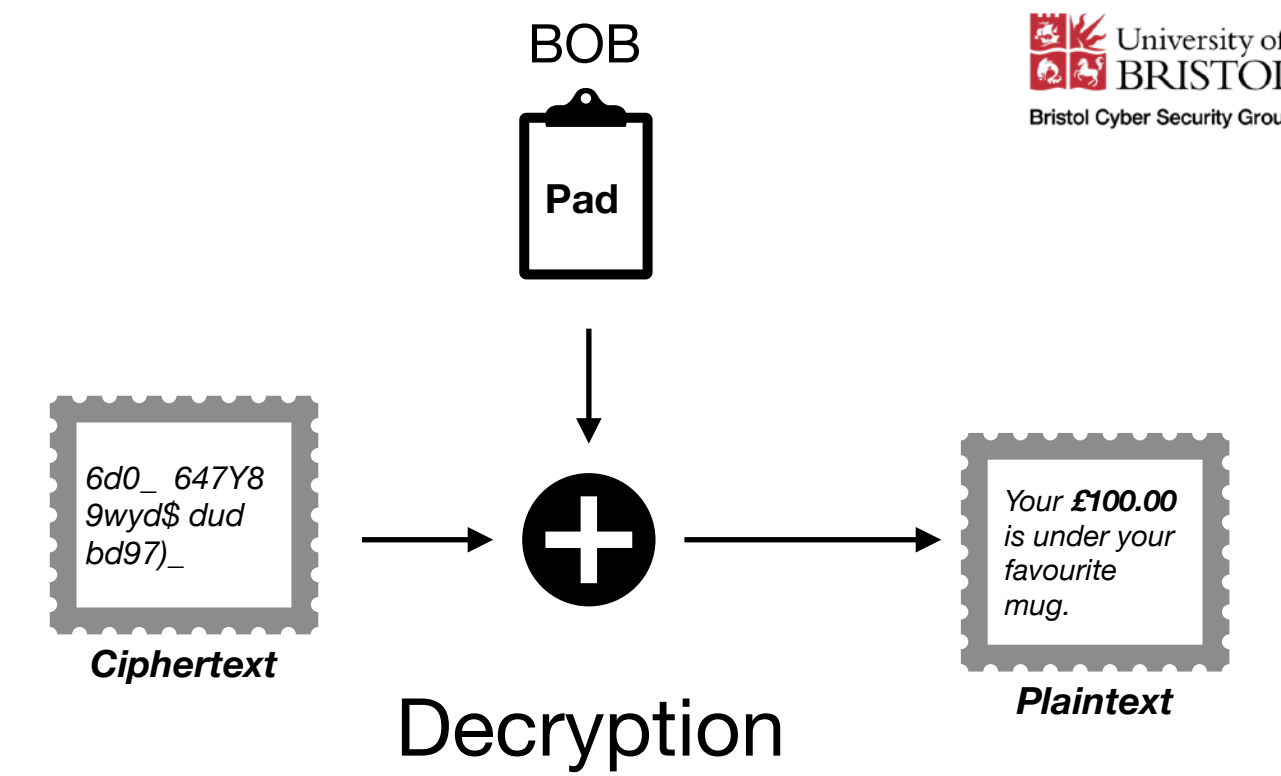
CRYPTOGRAPHIC ALGORITHMS - ONE-TIME PAD

To decrypt the message

$$M' \oplus K = M$$

$$\text{Key} = K * e 0 !$$

3 4F 9 5C 4E →
HEX



Cryptography

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Binary

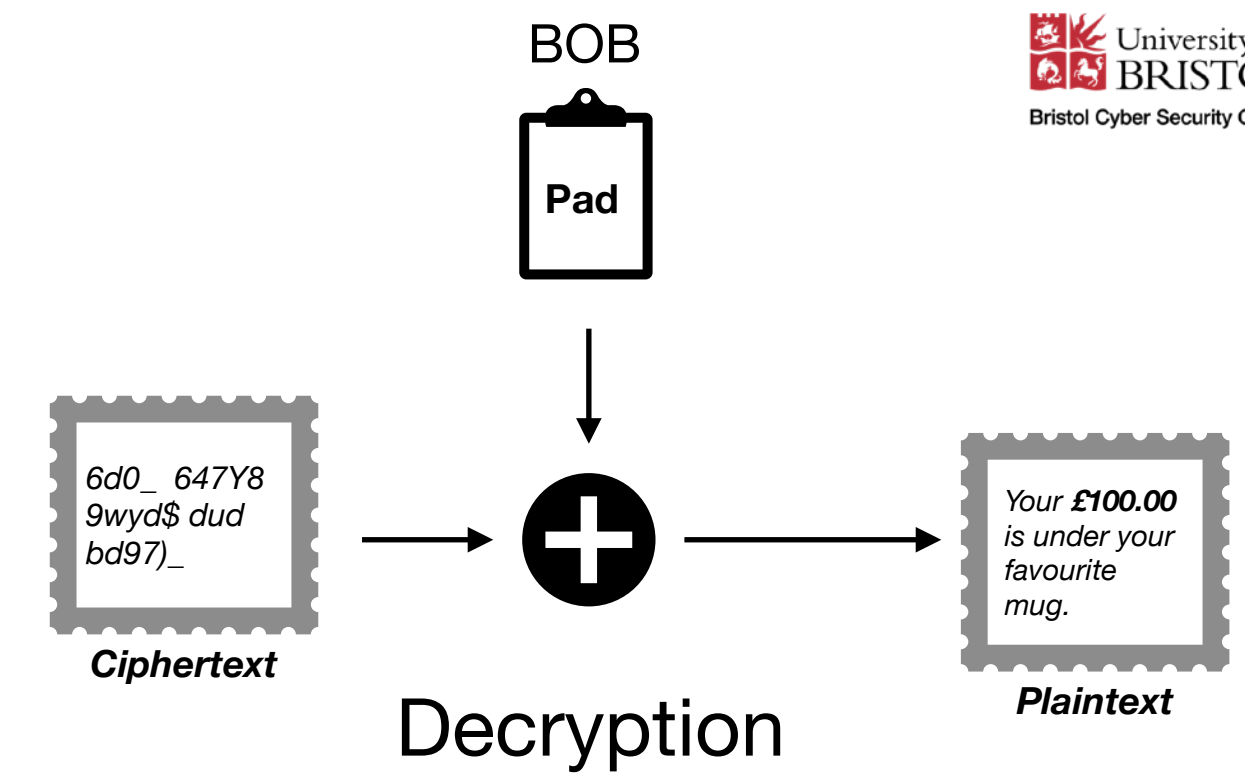
0000 0011

0100 1111

0000 1001

0101 1100

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Cryptography

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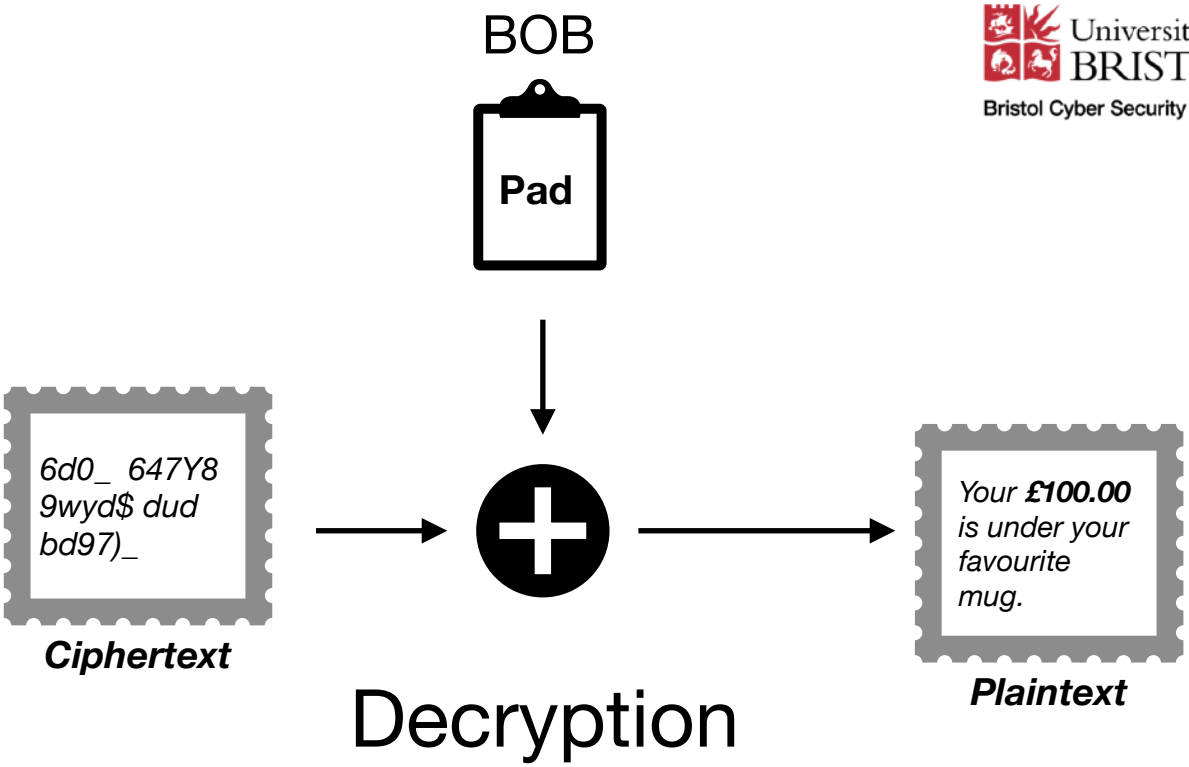
Key = K * e 0 !

Binary	
0000 0011	<u>0000 0011</u> =
0100 1111	<u>0100 1111</u> =
0000 1001	<u>0000 1001</u> =
0101 1100	<u>0101 1100</u> =
0100 1110	<u>0100 1110</u> =

3 4F 9 5C 4E

HEX

→



Cryptography

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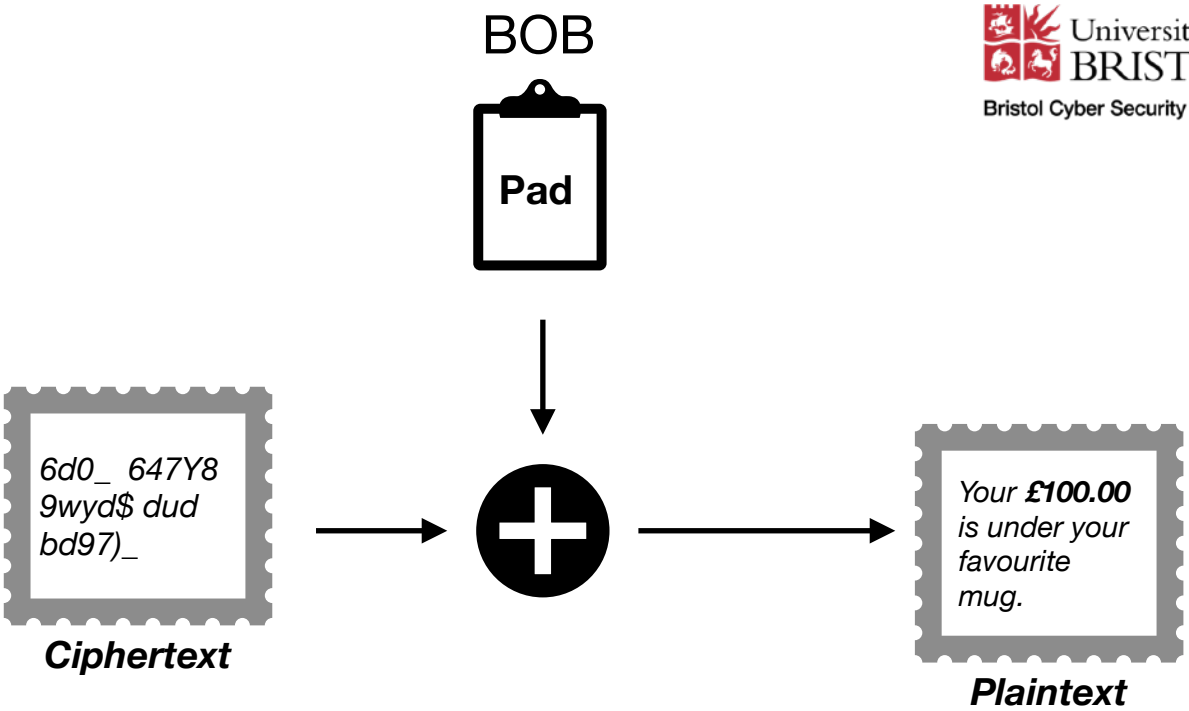
Key = K * e 0 !

Binary

0000 0011	K 0000 0011 0100 1011 =
0100 1111	* 0100 1111 0010 1010 =
0000 1001	e 0000 1001 0110 0101 =
0101 1100	0 0101 1100 0011 0000 =
0100 1110	! 0100 1110 0010 0001 =

3 4F 9 5C 4E →

HEX



Cryptography

CRYPTOGRAPHIC ALGORITHMS - ONE-TIME PAD

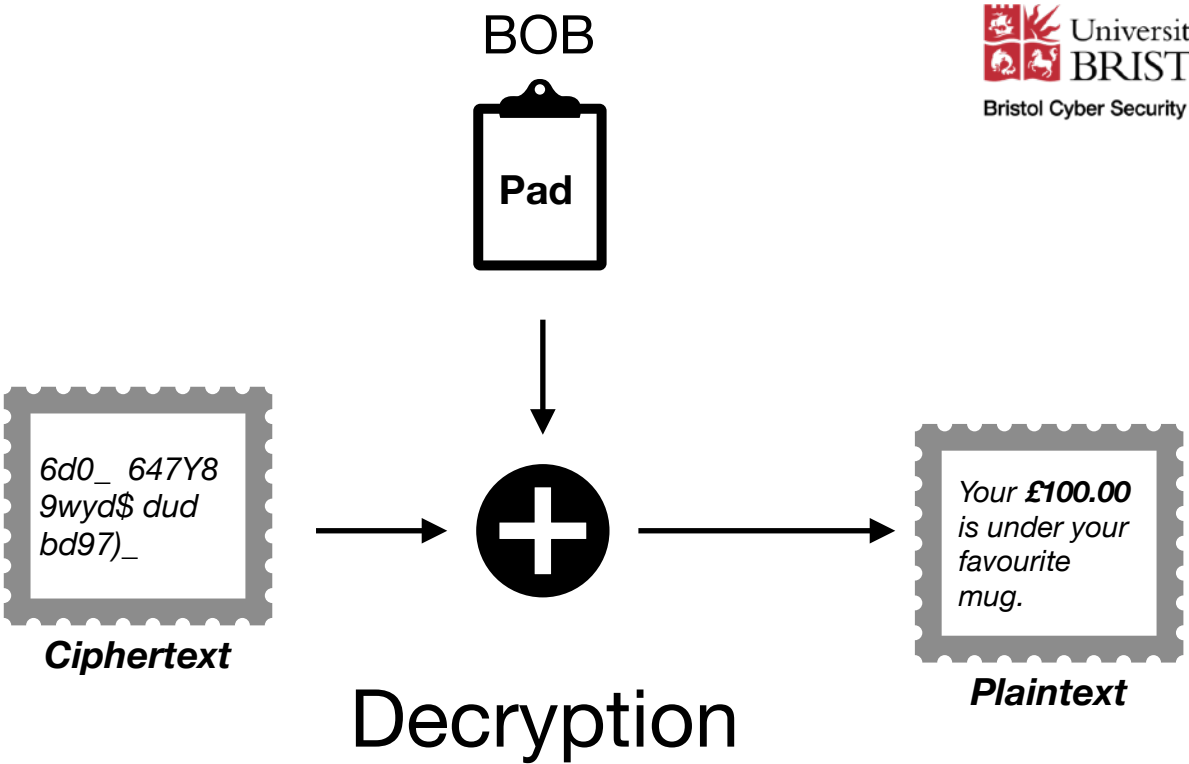
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$M' \oplus K = M$

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3 4F 9 5C 4E →
HEX

Binary	Results of XOR	
0000 0011	K $\frac{0000\ 0011}{0100\ 1011}$	= 0100 1000
0100 1111	* $\frac{0100\ 1111}{0010\ 1010}$	= 0110 0101
0000 1001	e $\frac{0000\ 1001}{0110\ 0101}$	= 0110 1100
0101 1100	0 $\frac{0101\ 1100}{0011\ 0000}$	= 0110 1100
0100 1110	! $\frac{0100\ 1110}{0010\ 0001}$	= 0110 1111



Cryptography

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3 4F 9 5C 4E
HEX

→

Binary

0000 0011

0100 1111

0000 1001

0101 1100

0100 1110

Results of XOR

K

0000 0011

0100 1011

= 0100 1000

*

0100 1111

0010 1010

= 0110 0101

e

0000 1001

0110 0101

= 0110 1100

o

0101 1100

0011 0000

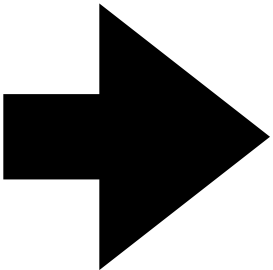
= 0110 1100

!

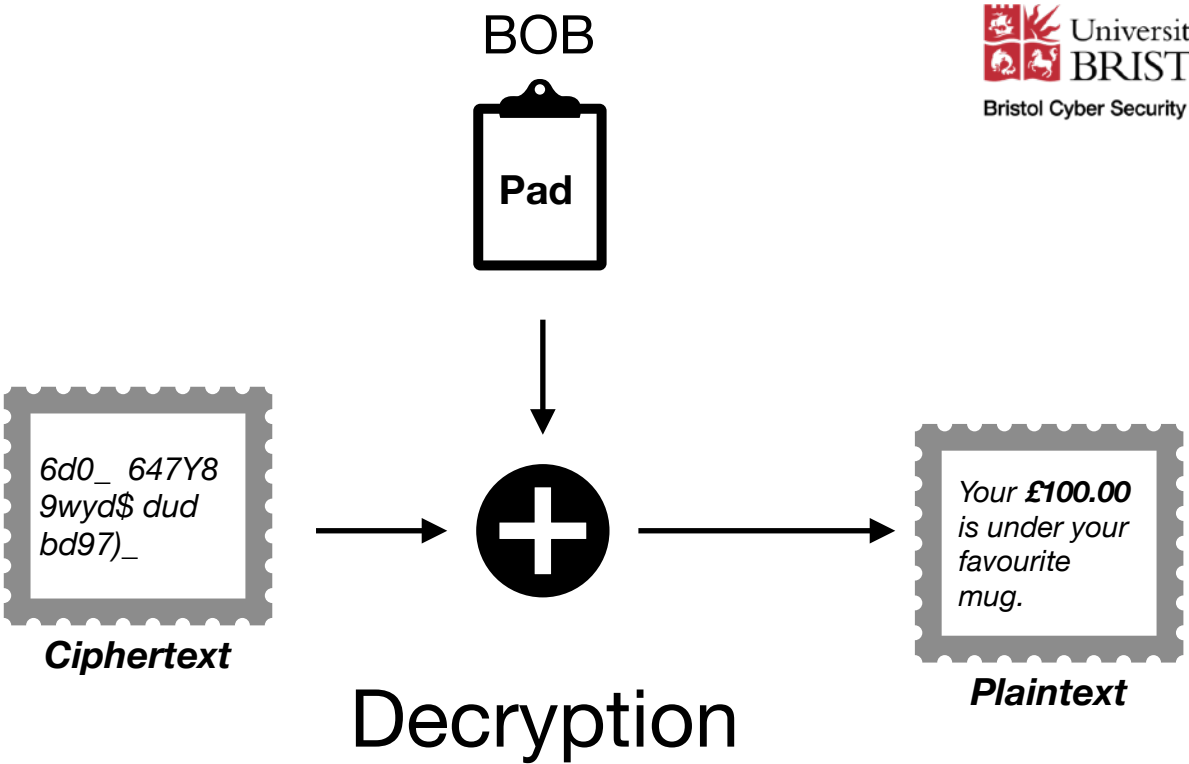
0100 1110

0010 0001

= 0110 1111



H	1001000
e	1100101
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Cryptography

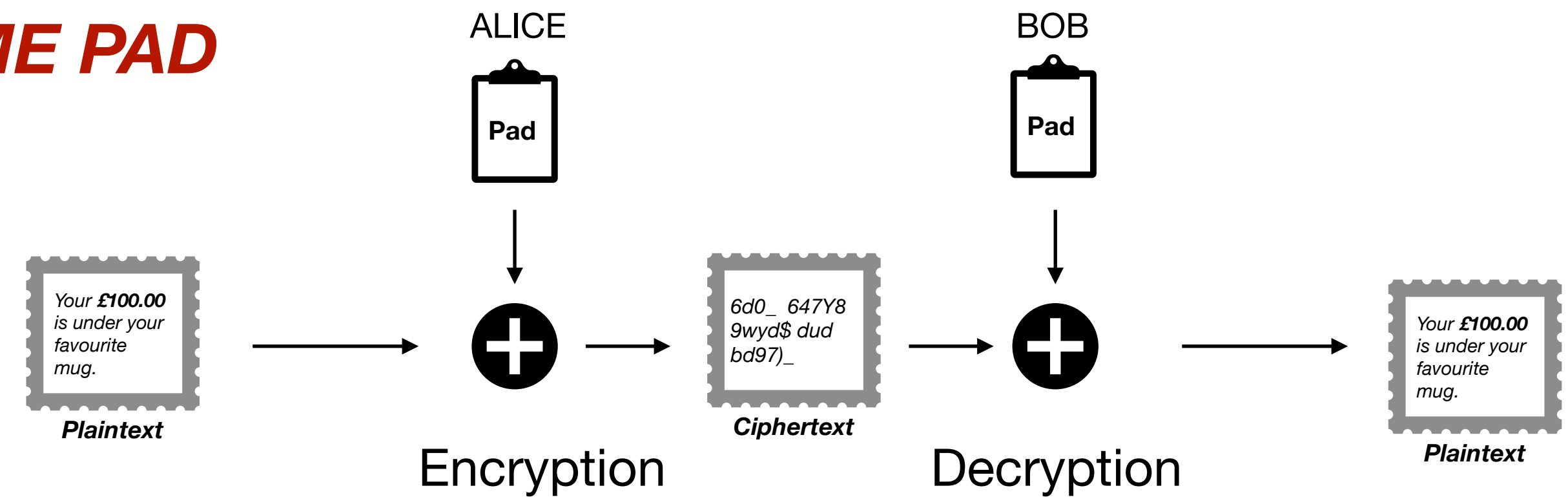
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Key = K * e 0 !

Message = H E L L O

Message = C L A S S

Key = 7 T * 2 e 6 !

Message = \$100.00

Message = P l a n e t s

Key = Q 7 7 T * 2 e 6 ! c \$ i

Message = One-Time-pad

Key = K * e 0 !

Ciphertext (HEX): 3 4F 9 5C 4E

Key = HCUEGDQRTYQ

Cipher (Char): AJCWOVQKXQJ

Cryptography

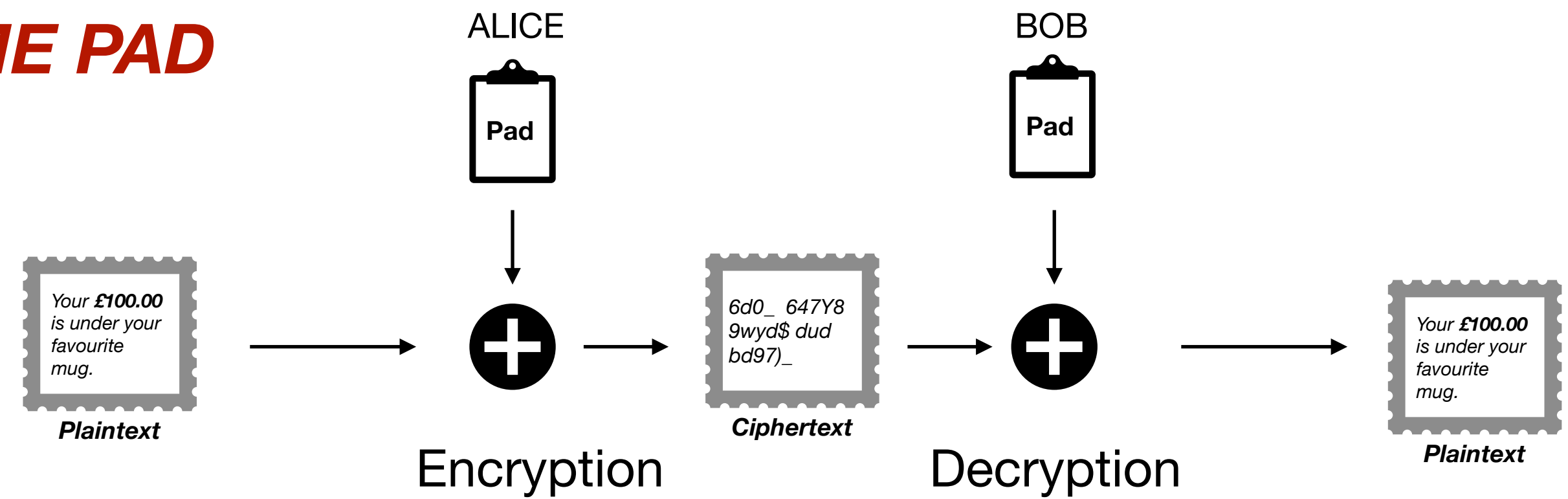
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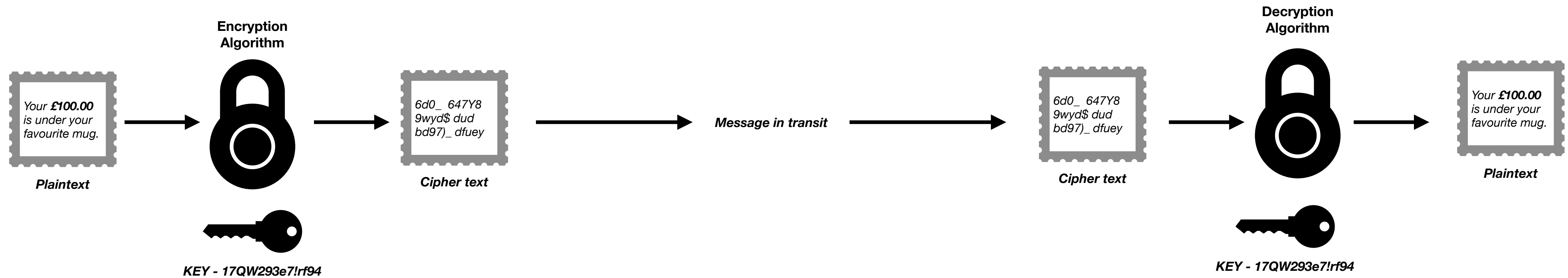
Two wrong and right

- One time pad cipher is theoretical considered the most secure cipher.
- As long as the pad is kept secret, it can be reused several times.
- The length of the does not necessarily need to be the same length as the message.

Cryptography

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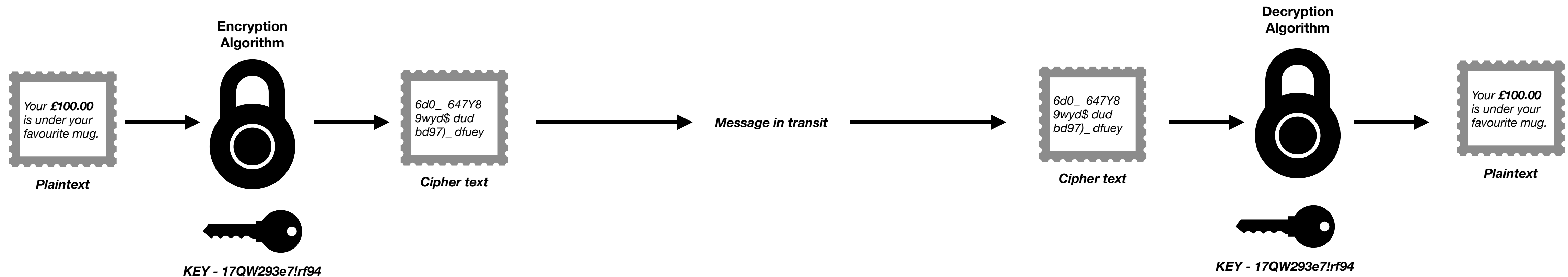
Data Encryption Standard (DES)



Cryptography

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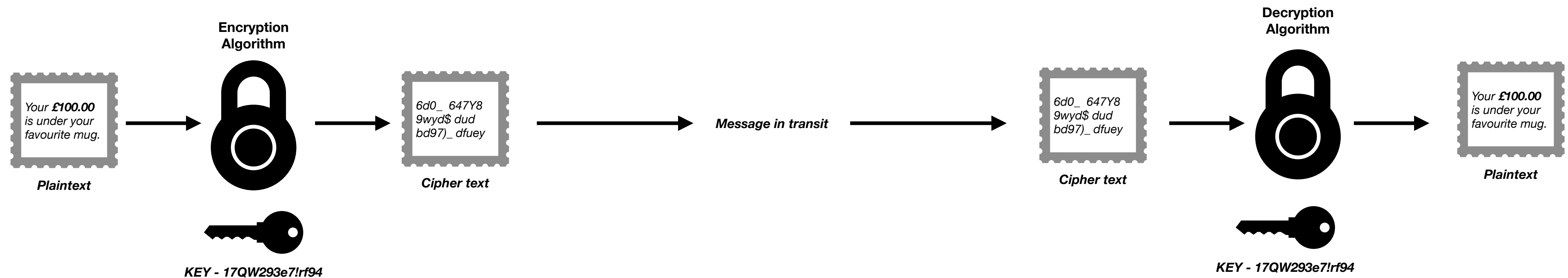


- A widely used algorithm
- DES designed by IBM in 1977
- 56 bits keys
- Ciphering of 64 bits blocks
- Based on the permutation mechanisms and XOR over keys

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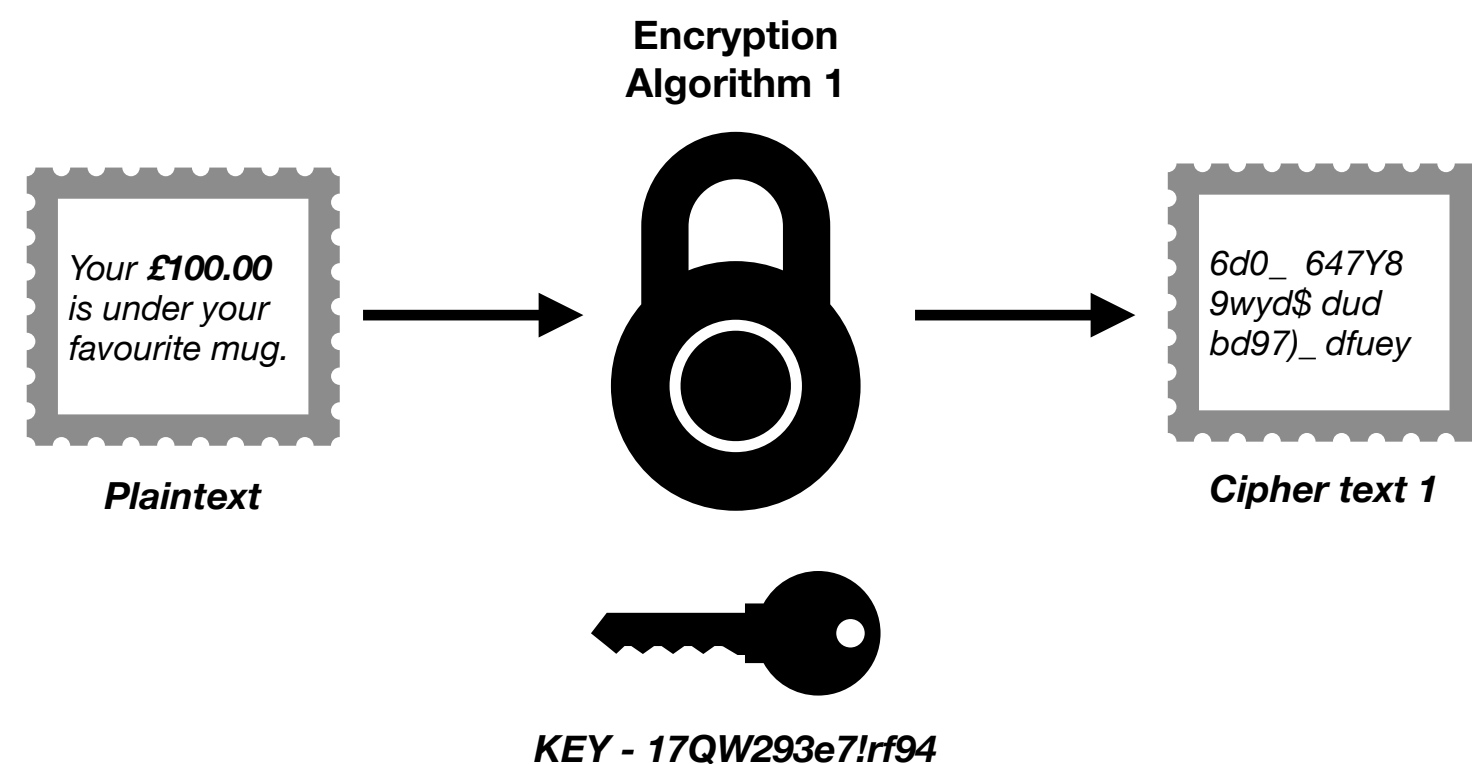


- A widely used algorithm
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- 56 bits keys
- Ciphering of 64 bits blocks
- Based on the permutation mechanisms and XOR over keys
- DES cracking attacks: The original DES cipher's key size of 56 bits was generally sufficient when that algorithm was designed, but the availability of increasing computational power made brute-force attacks feasible.
 - Brute force attacks (testing the 256 keys possibilities)
 - 18 January 1999, Electronic Frontier Foundation, 100000 PC, 22h15

Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

Triple Data Encryption Standard (3DES)



3DES was designed to replace DES without the need to design a completely new block cipher.

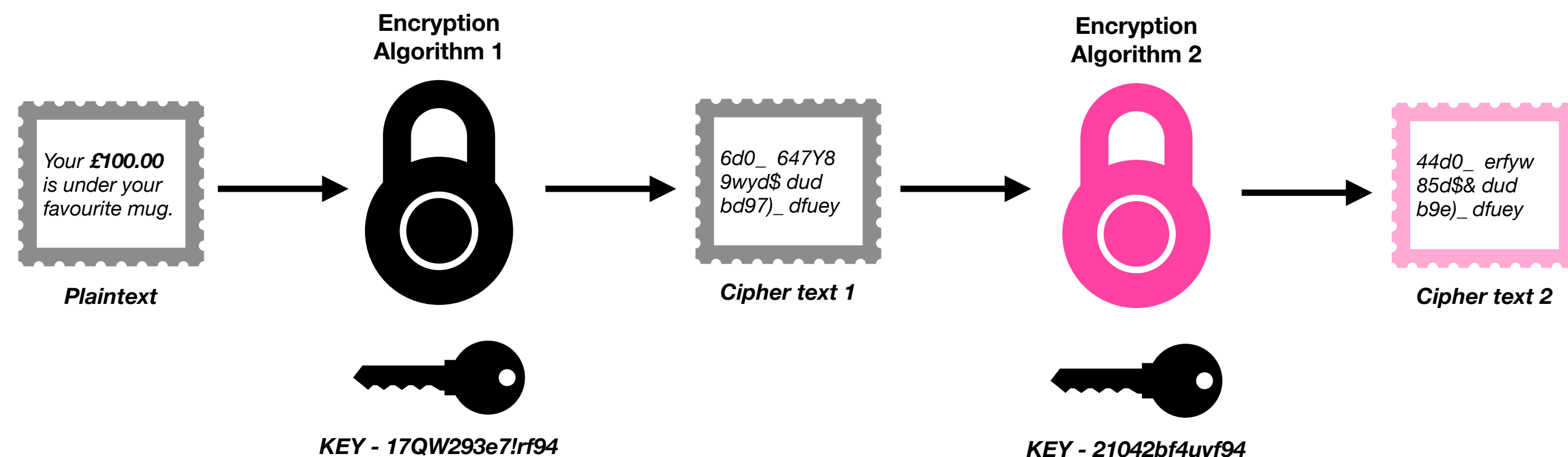
As its name implies, 3DES uses **three rounds** of encryption instead of just one. The **ciphertext** of one round becomes the entire input for the second iteration. 3DES employs a total of 48 iterations in its encryption (3 iterations x 16 rounds). The most secure versions of 3DES use different keys for each round.

- A naive approach to increase strength of a block encryption algorithm with short key length (like DES) would be to use two keys (K1, K2) instead of one, and encrypt each block twice: $E_{K2}(E_{K1}(\text{plaintext}))$.
- If the original key length is n bits, one would hope this scheme provides security equivalent to using key $2n$ bits long.
- Unfortunately, this approach is vulnerable to man-in-the-middle attack:
 - Given a known plaintext pair (x, y) , such that $y = E_{K2}(E_{K1}(x))$, one can recover the key pair (K1, K2) in $\sim 2n$ steps, instead of $\sim 2^{2n}$ steps one would expect from algorithm with $2n$ bits of key.

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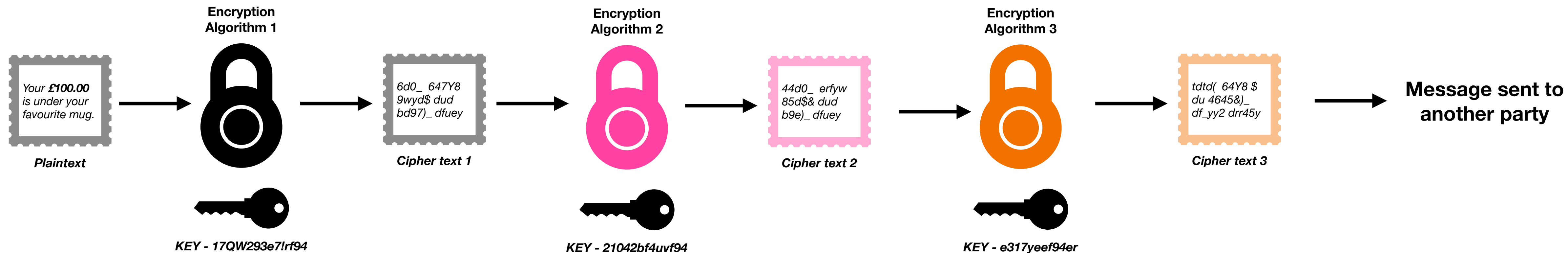
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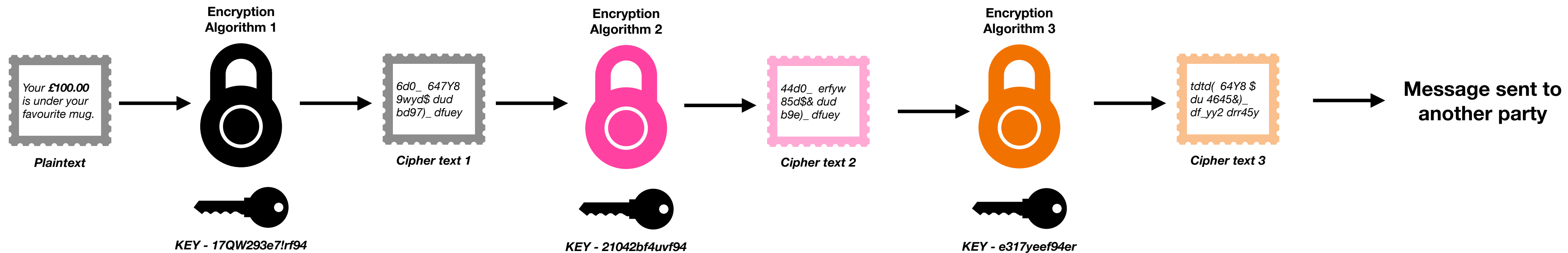
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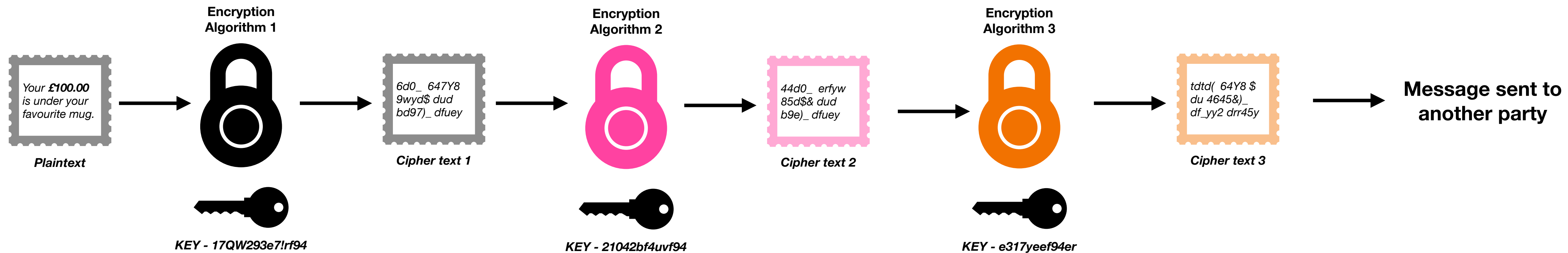


- Triple DES uses a "key bundle" that comprises three DES keys, K1, K2 and K3, each of 56 bits (excluding parity bits).
- The encryption algorithm is:
Ciphertext = **EK1**(plaintext)

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Triple Data Encryption Standard (3DES)

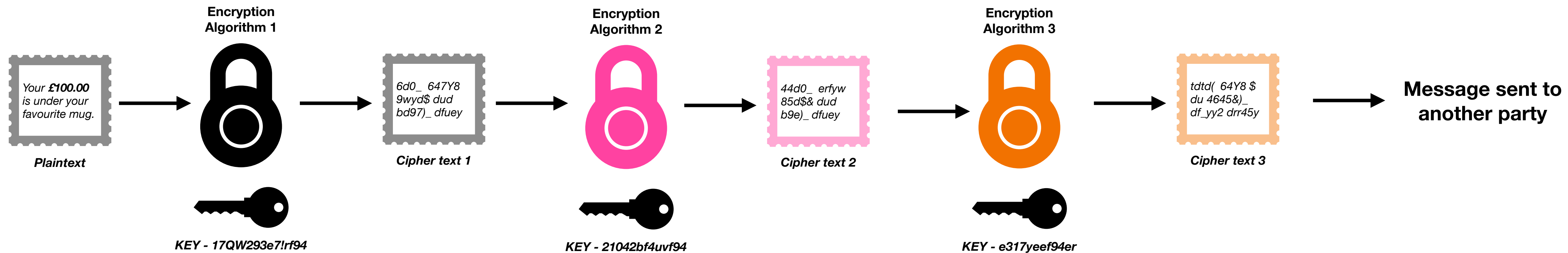


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Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

Triple Data Encryption Standard (3DES)

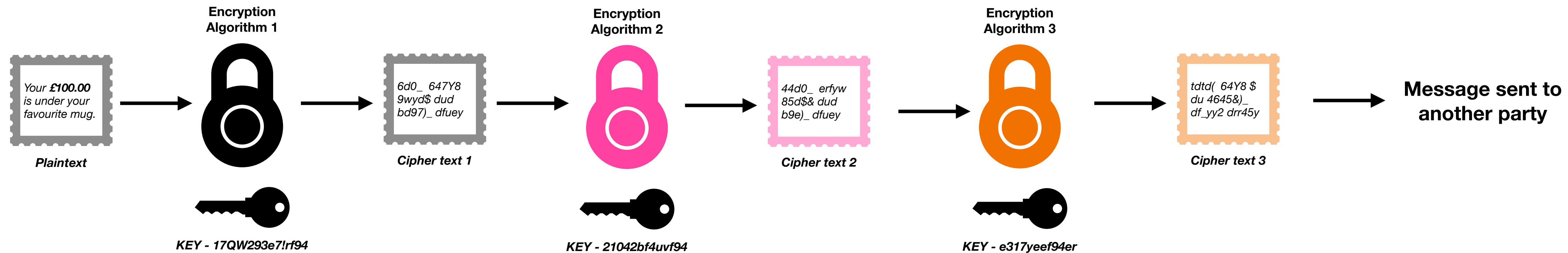


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Cryptography

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Triple Data Encryption Standard (3DES)



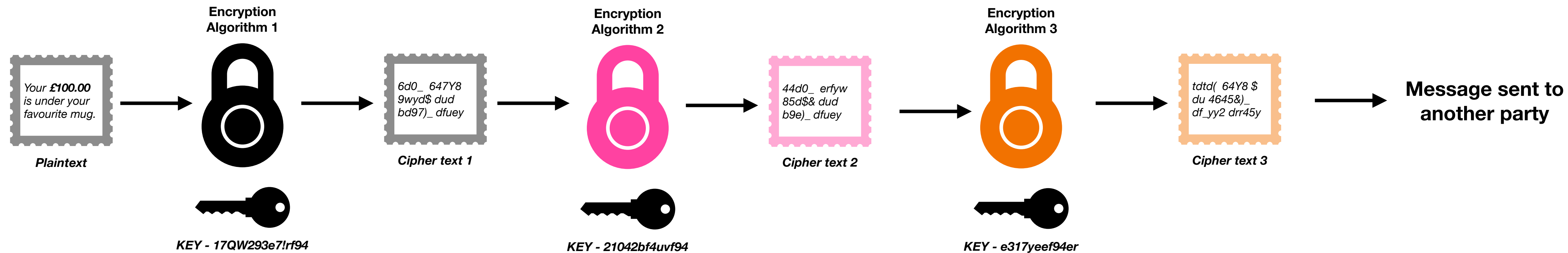
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$$\text{Ciphertext} = \text{EK}_3(\text{DK}_2(\text{EK}_1(\text{plaintext}))),$$
 i.e., DES encrypt with K1, DES decrypt with K2, then DES encrypt with K3.

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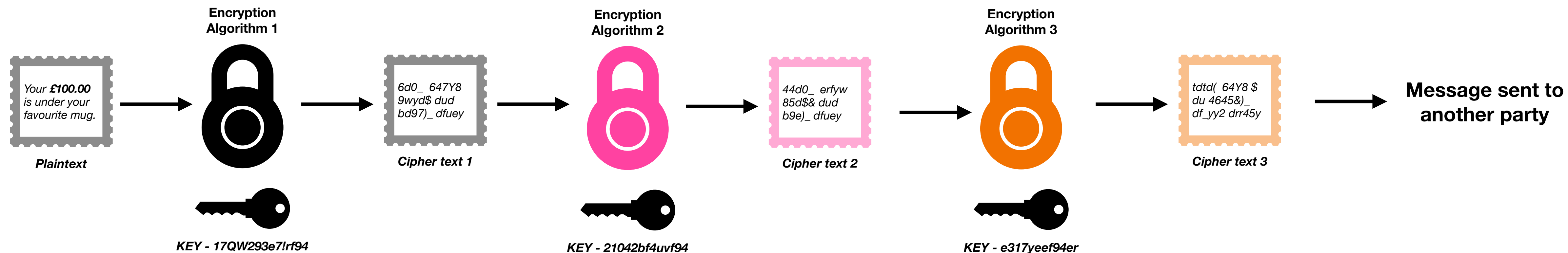
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- Decryption is the reverse:

$$\text{Plaintext} = \text{DK}_3(\text{cipher text})$$

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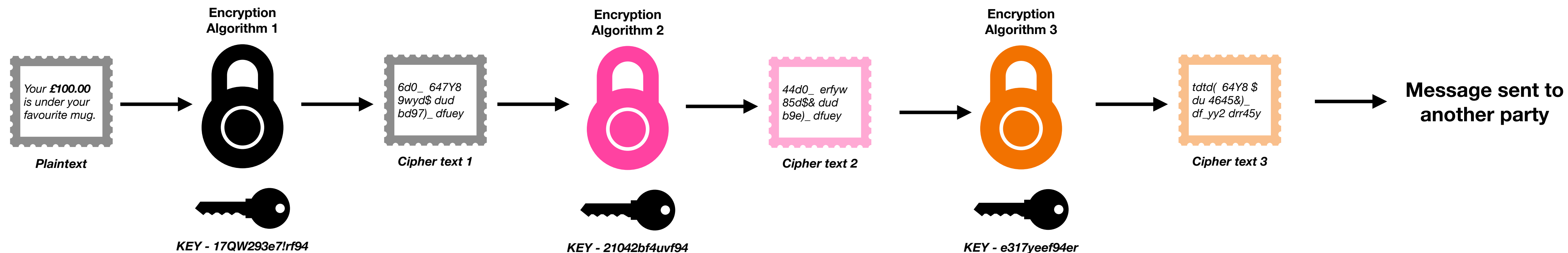
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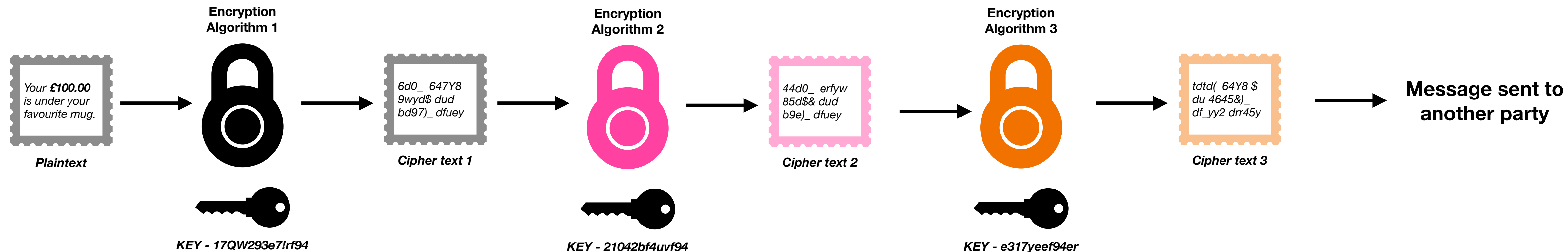
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Cryptography

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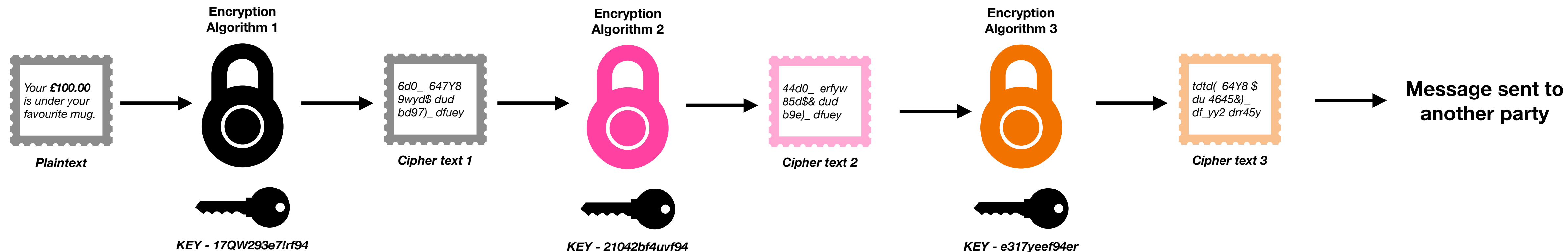
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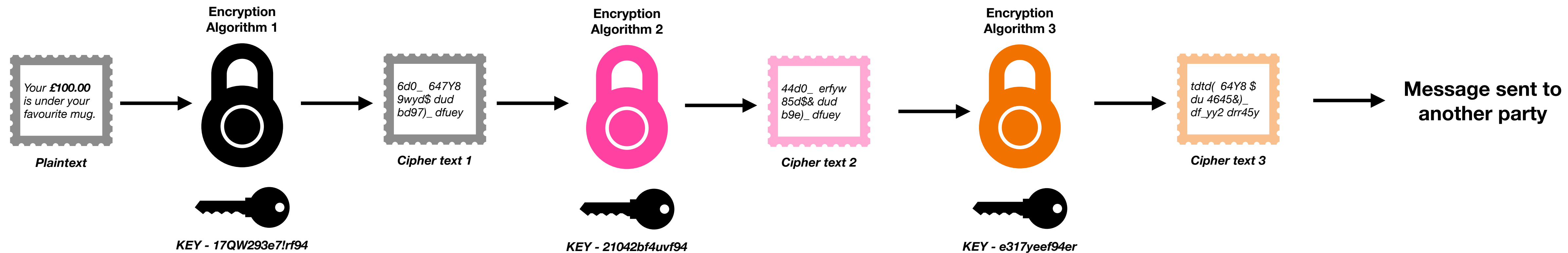
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- Each triple encryption encrypts one block of 64 bits of data.
- In each case the middle operation is the reverse of the first and last. This improves the strength of the algorithm when using keying option 2, and provides backward compatibility with DES with keying option 3.

Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

Triple Data Encryption Standard (3DES)



Two wrong and right

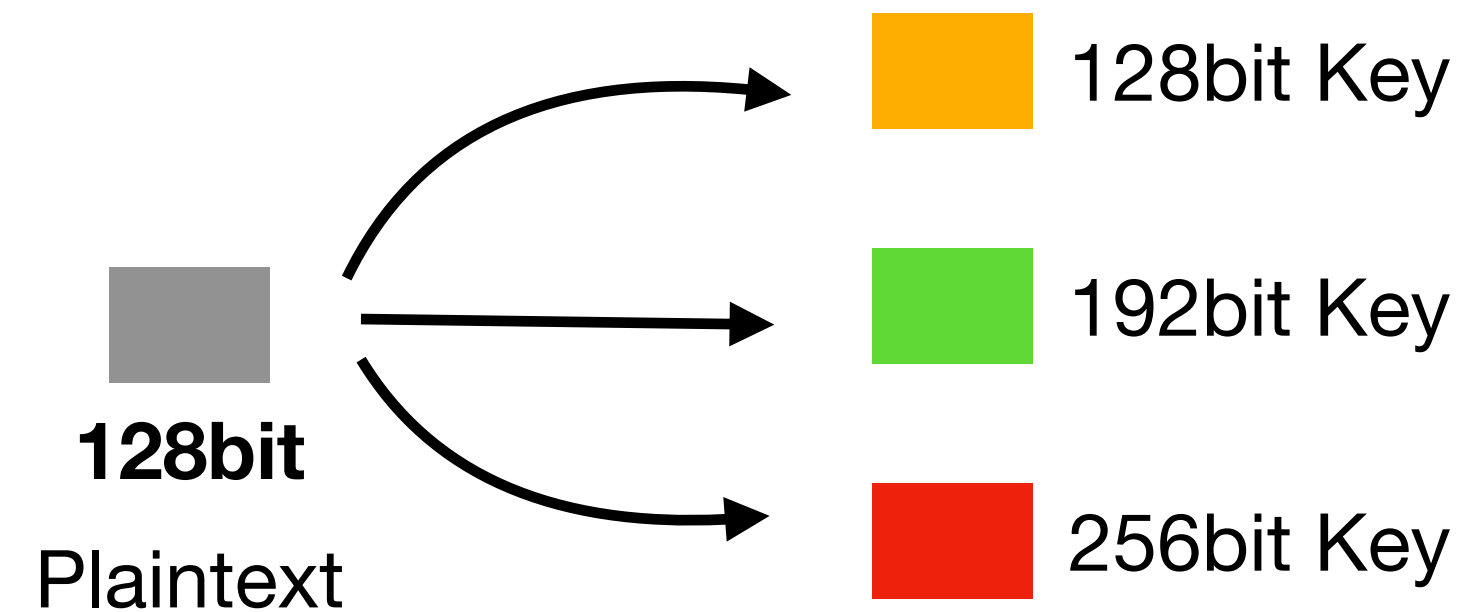
- Triple DES uses three different keys to encrypt the messages.
- The first ciphertext should always be the same as the last ciphertext
- Triple DES is weaker than the DES because it repeats the encryption of the ciphertext.

Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

Advanced Encryption Standard (AES)

The Advanced Encryption Standard (AES) is a symmetric algorithm that performs three steps on every block (128 bits) of plaintext. Within step 2, multiple rounds are performed depending upon the key size: a 128-bit key performs ten rounds, a 192-bit key performs 12 rounds, and a 256-bit key, known as AES-256, uses 14 rounds. Within each round, bytes are substituted and rearranged, and then special multiplication is performed based on the new arrangement. To date, no attacks have been successful against AES.

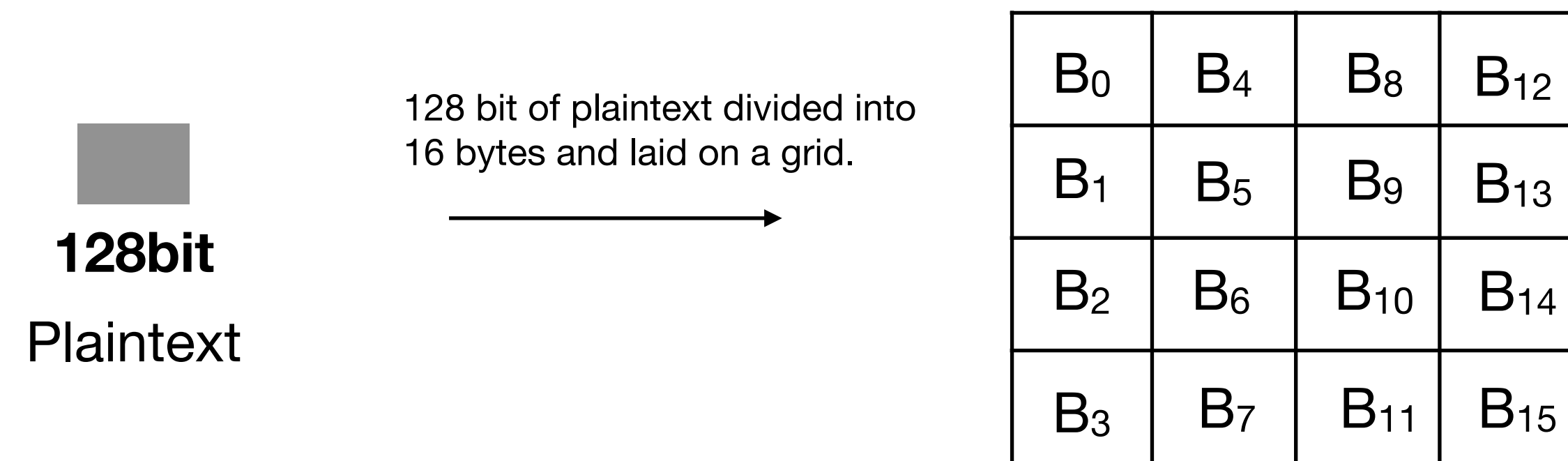


128 bit of plaintext can either be encrypted by 128bit key or 192bit key or 256bit key.

Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

Advanced Encryption Standard (AES)



Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

Advanced Encryption Standard (AES)


128bit
Plaintext

128 bit of plaintext divided
into 16 bytes and laid on a
grid.



B ₀	B ₄	B ₈	B ₁₂
B ₁	B ₅	B ₉	B ₁₃
B ₂	B ₆	B ₁₀	B ₁₄
B ₃	B ₇	B ₁₁	B ₁₅

Few operations that AES would do, everything
happens in the grid.

1. Use our part of the key perform XoR operation
2. Substitution Bytes
3. Shift rows
4. Mix columns
5. Add another part of the key

} Complete
Round

Cryptography

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} Permutation

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Remember:

128bit key performs 10 rounds

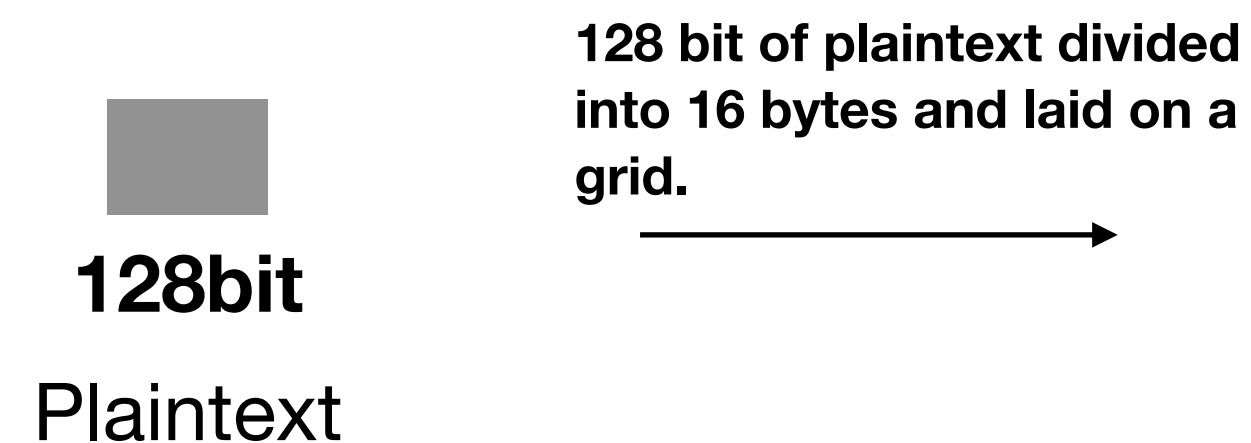
192bit key performs 12 rounds

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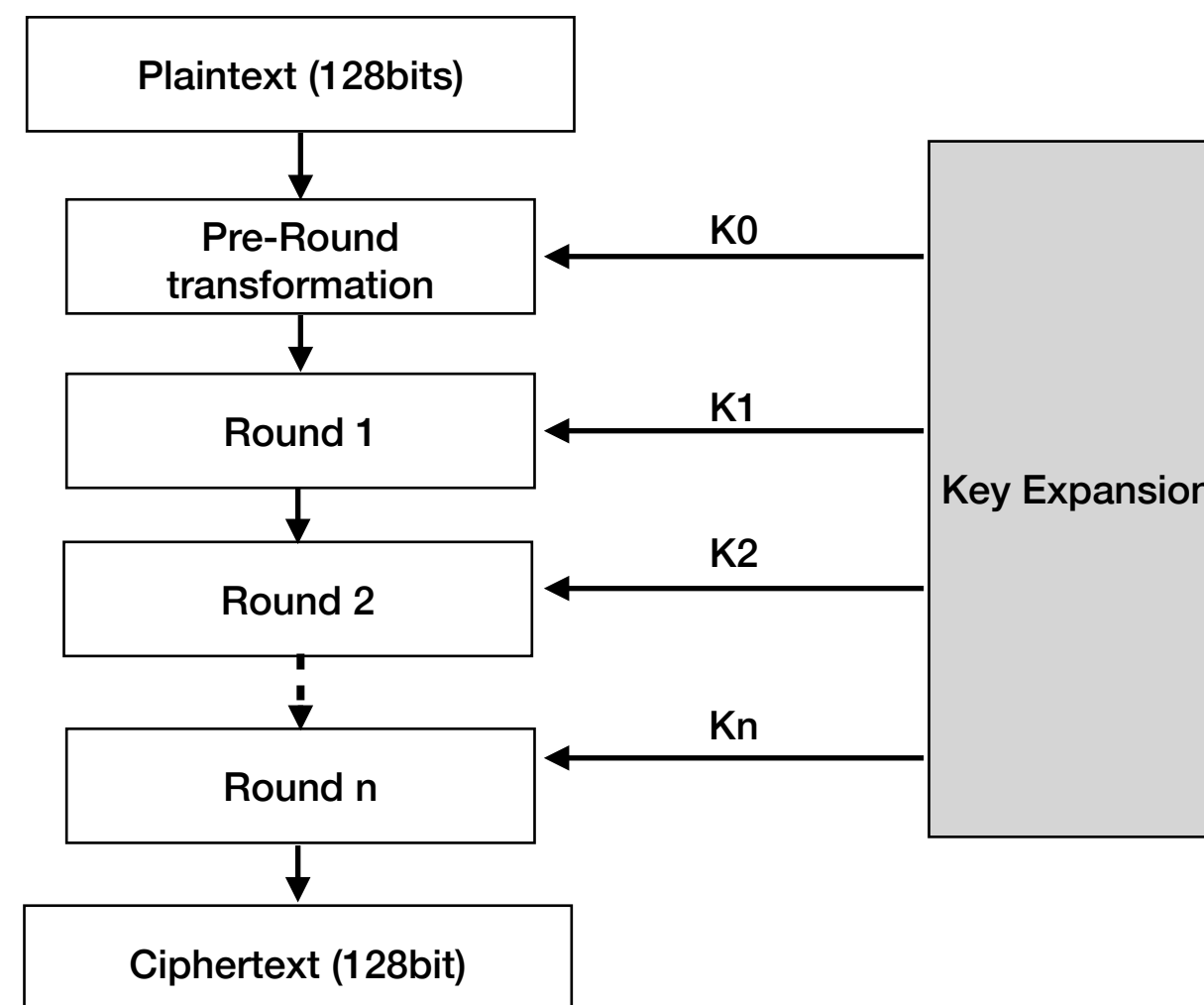
Complete Round

Remember:

128bit key performs 10 rounds

192bit key performs 12 rounds

256bit key performs 14 rounds

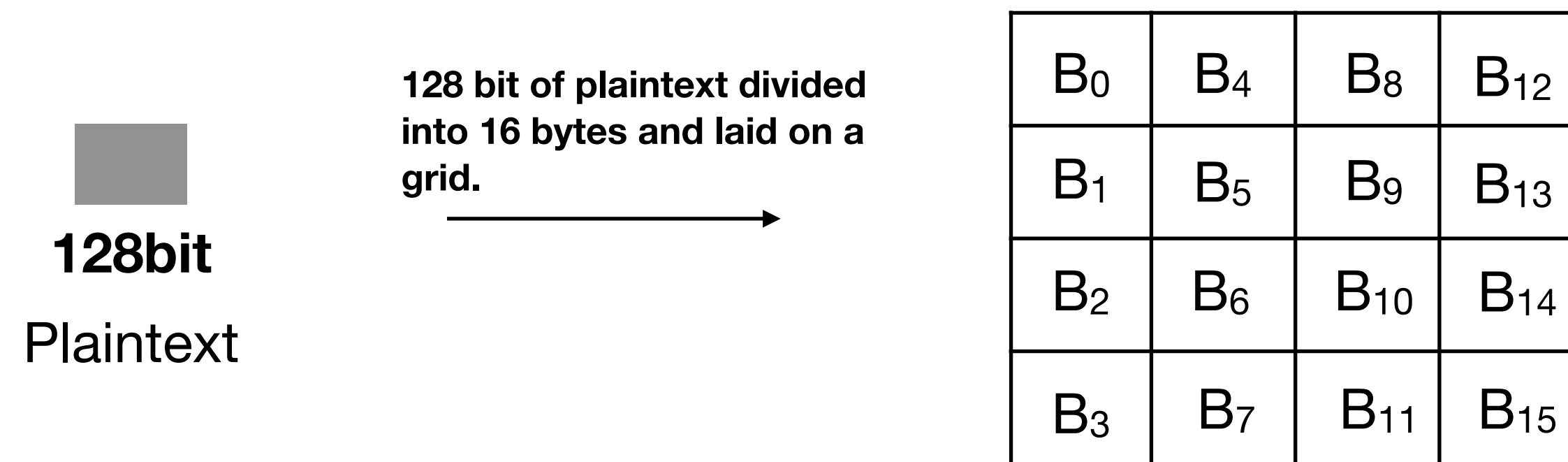


The key is first divided into several parts depending on the number of rounds, so that for each round the key is different.

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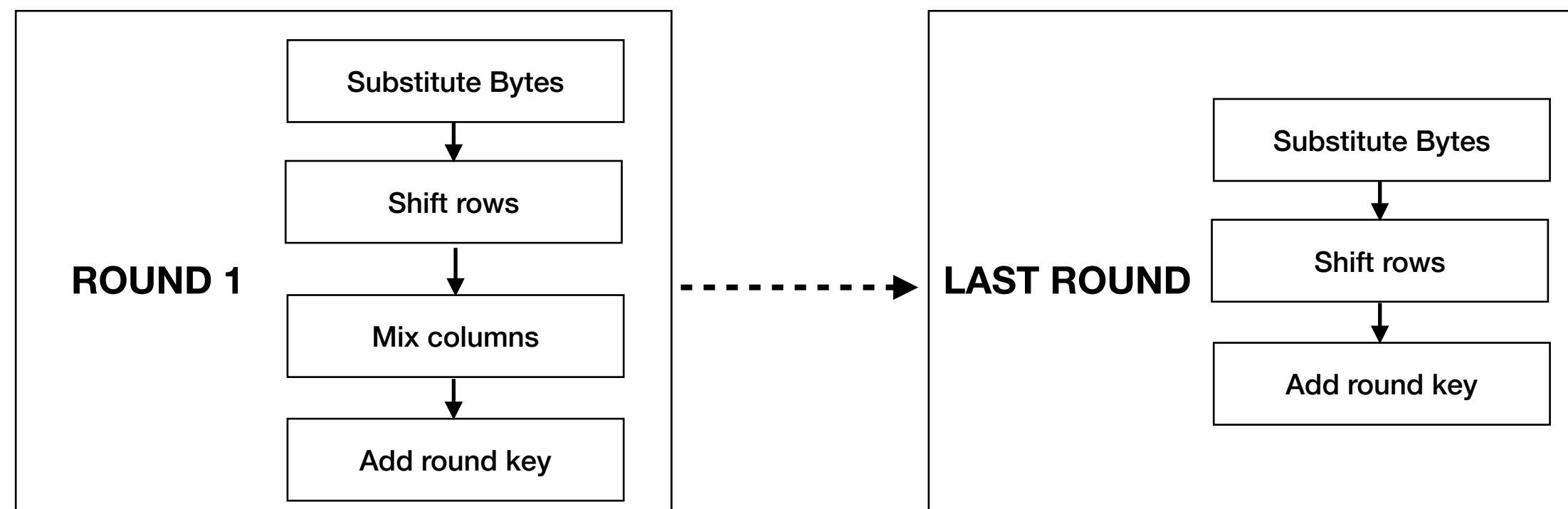
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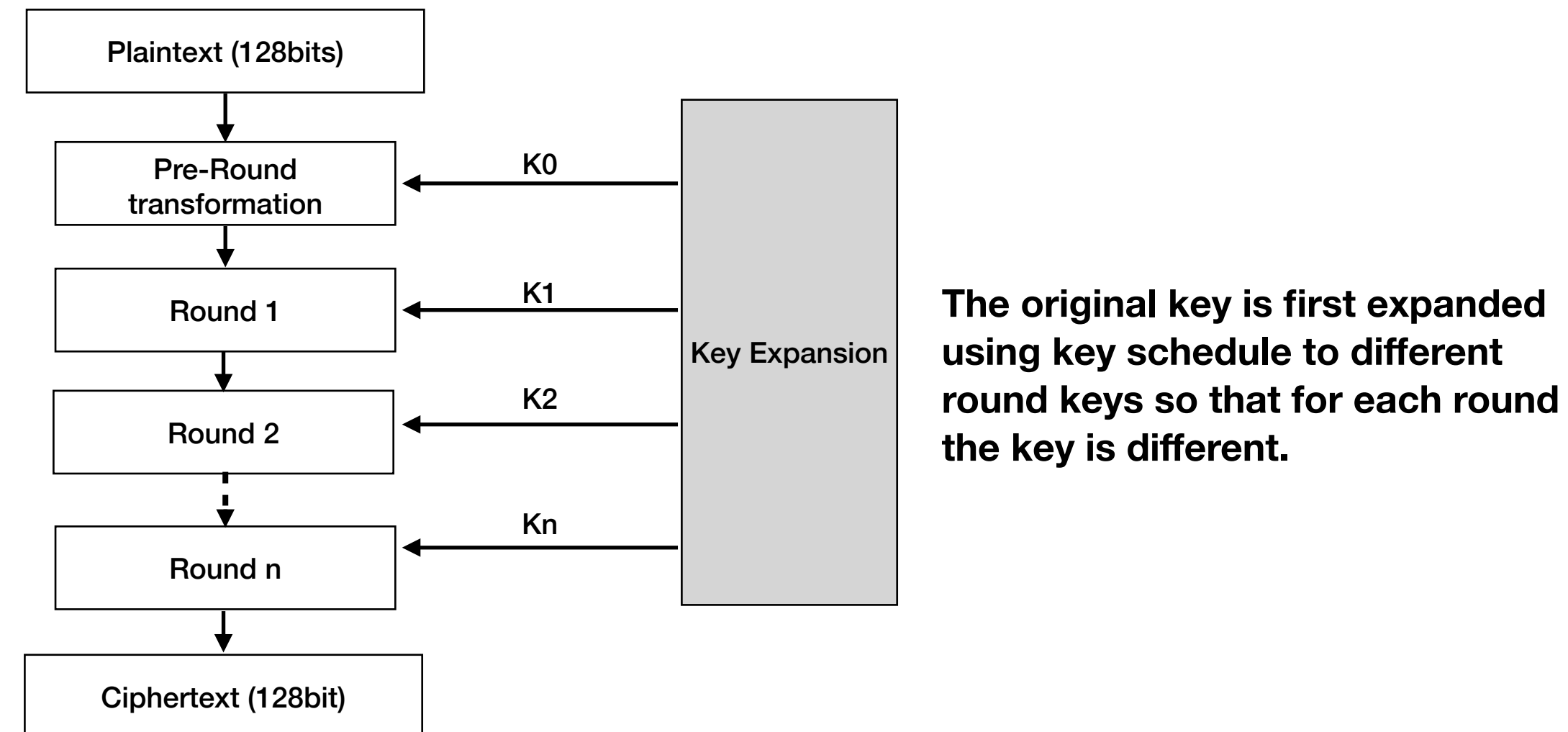
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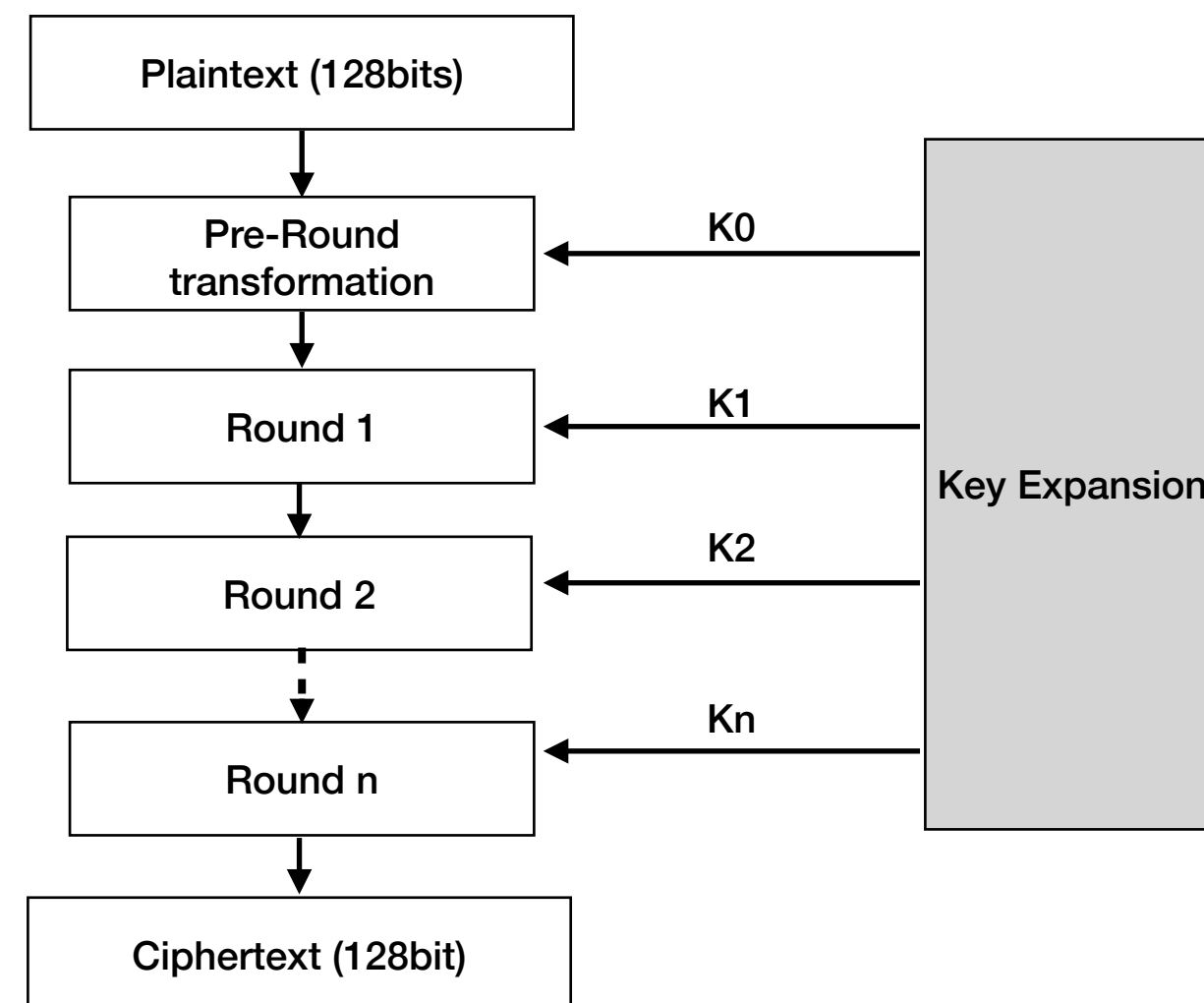
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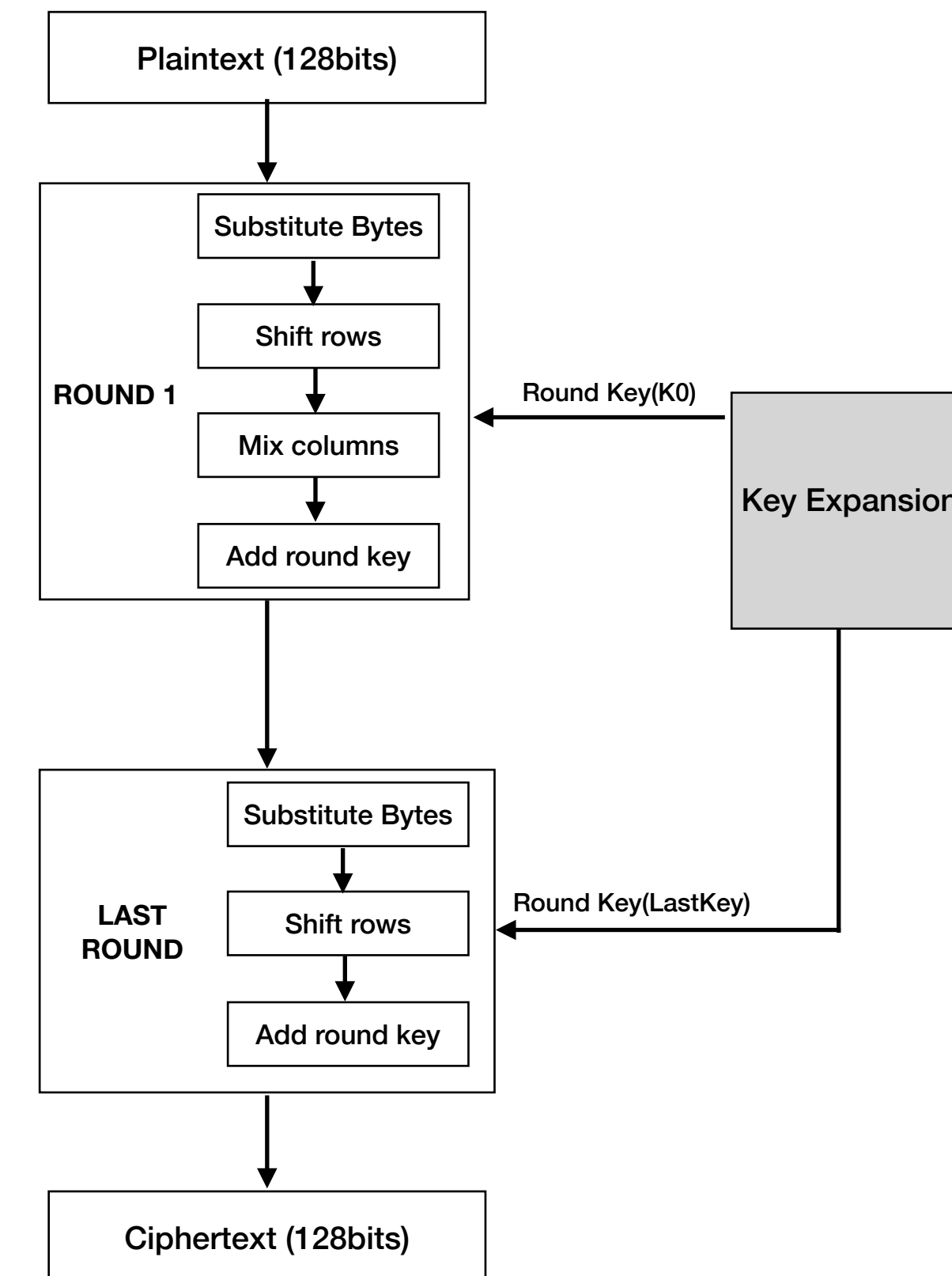
Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

Advanced Encryption Standard (AES)



The original key is first expanded using key schedule to different round keys so that for each round the key is different.



Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

RIVEST CHIPHER (RC)

The RC algorithms are a set of symmetric-key encryption algorithms invented by Ron Rivest. Despite the similarity in their names, the algorithms are for the most part unrelated. There have been six RC algorithms so far:

- RC1 was never published.
- RC2 was a 64-bit block cipher developed in 1987.
- RC3 was broken before ever being used.
- RC4 is the world's most widely used stream cipher.
- RC5 is a 32/64/128-bit block cipher developed in 1994.
- RC6, a 128-bit block cipher based heavily on RC5^[2], was an AES finalist developed in 1997.

Cryptography

CRYPTOGRAPHIC ALGORITHMS - SYMMETRIC ENCRYPTION

BLOWFISH

Blowfish is a block cipher algorithm that operates on 64-bit blocks and can have a key length from 32 to 448 bits.

It was created by Bruce Schneier in 1993. Symmetric encryption uses a single encryption key to both encrypt and decrypt data. The sensitive data and the symmetric encryption key are utilized within the encryption algorithm to turn the sensitive data into ciphertext. Blowfish, along with its successor Twofish, was in the running to replace the Data Encryption Standard (DES) but failed due to the small size of its block. Blowfish uses a block size of 64, which is considered wholly insecure. Twofish fixed this issue, by implementing a block with a size of 128. Blowfish is much faster than DES, but it trades in its speed for security.